/s/peaking while Trans: Transfeminine Navigations of the Cisgender Listener

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List of Abbreviations

CoG	Spectral Center of Gravity; an acoustic measure, in Hz, of /s/ frontness (higher CoG = fronter /s/). CoG is calculated as the mean of the frequencies of a sound at a given moment, weighted by amplitude.
CP	Conversation Partner; the term I use to describe the cisgender woman and transgender woman who played the role of the other side of each conversation the speaker participants had.
HRT	Hormone Replacement Therapy; a common medical intervention for gender transition.
BIPoC	Black, Indigenous, and People of Color.

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Abstract

Sociolinguistic research on transgender speech has primarily been of two categories: aggregated comparisons against cisgender speakers, and ethnographic exploration within queer in-groups; neither of these address how trans speakers navigate cisgender perceptions. Frontness of the /s/ sound is a well-studied index of femininity and gay-masculinity within the gender norms of the cisgender white English-speaking population; I investigate how four transfeminine speakers navigate these norms through a comparison of their /s/ pronunciations with two conversation partners, one transgender woman and one cisgender woman, to fill this gap. I audio-recorded one-on-one casual conversations between each speaker and each conversation partner, acoustically comparing speakers' pronunciations of /s/ between the two conversations. I then interviewed each speaker, assessing metalinguistic concerns and attitudes in three areas hypothesized to predict shifts: speakers concerned with cisgender perceptions were expected to defensively shift towards hegemonically normative feminine /s/ pronunciations, whereas speakers who were concerned with sounding gay-masculine or who distanced themselves from hegemonic femininity were expected not to. I find that two speakers shifted towards fronter /s/ pronunciations with the cis listener, and two speakers did not. Counter to my predictions, these speakers who shifted to fronter /s/ pronunciations were the only two to express concerns about sounding like gay men in the interviews, whereas all speakers were similarly concerned with cis perceptions, and all speakers except for one, who did not shift to fronter /s/ pronunciations, distanced themselves from hegemonic femininity. Reinterpreting my initial model, I consider /s/ fronting to hold greater salience for speakers concerned with sounding gay-masculine – both as an index of gay-masculinity and as an index of femininity – highlighting the interdependence of these indexical orders. These speakers take up the feminine meaning of /s/ fronting as a resource for self-defense against potentially-misgendering cisgender listening practices. I conclude that marginalized speakers make use of complex indexical understandings to navigate normative systems, and call for further research on the mediation between speech production and speech perception.

Introduction

Along with the visual presentation of one's body, the sound of one's voice is among the most strongly gendered aspects of behavior. Transgender people[1] know this intimately, as our speech is highly scrutinized by those who – consciously or unconsciously – seek to enforce gender normativity (Jessee and Calder 2025). In this thesis, I investigate how this knowledge manifests by comparing English-speaking transfeminine[2] speakers' linguistic behavior across two conversational contexts: a conversation with a cisgender[3] woman, and a conversation with a transgender woman. I focus on their pronunciation of the /s/[4] sound (the first sound in 'safe' and the last sound in 'face'), a well-studied site of gendered linguistic variation.

It has become widely accepted both in the academic study of gender and parts of the popular consciousness that gender is performative (Butler 1990; Henry and Hall 2025). Most simply, this means that gender does not exist outside of cultural and historical context, and is constantly constructed and reconstructed by social behavior. Under this view, scholars of gender develop their models by studying the behaviors involved in gender, and how those behaviors come to have gendering effects. Language is of primary importance to the construction of gender; speech about gender is arguably the foundation of gender's existence. For instance, the speech of medical specialists conducting an ultrasound or delivering a newborn – "it's a girl!" or "it's a boy!", or the imperatives "act ladylike" and "man up", do the performative work of enforcing and solidifying gender (see Livia and Hall 1997b).

Yet, talk about gender does not represent the beginning and end of the linguistic performativity of gender. Sociolinguistics, the study of language and the social world, recognizes that variations in language use beyond the referential – less "what one says", more "how one says it" – are socially meaningful and per-

[1] I use 'transgender' or 'trans' to refer to those who identify with this label. Many transgender people understand the term broadly to include all who do not fully identify with the gender assigned to them at birth. However, the term is sometimes narrowed to refer mostly to transgender men and transgender women, who primarily identify with the opposite binary gender of the gender assigned to them at birth.

[2] I use the term 'transfeminine' to refer to those who identify with this label, primarily trans women and feminine nonbinary people (especially those assigned male at birth, for whom transition is a move towards a feminine identity).

[3] I use 'cisgender' or 'cis' to refer to people who are not transgender.

[4] In this thesis, slashes denote that the enclosed text is written in the International Phonetic Alphabet (see International Phonetic Association 1999). /s/ refers to the voiceless alveolar fricative speech sound. Although it looks the same as the English letter 's', this symbol refers to the sound, rather than the spelling.

formative (Henry and Hall 2025). The common belief that one's gender can be guessed or determined from the sound of their voice attests to the importance of the sound of the voice in the construction and performance of gender. For example, if a perceiver hears a voice that is higher pitched, they may assume the speaker is a woman, whereas if it is lower pitched, they may assume the speaker is a man. Here, the perceiver construes high pitch to be a sign of femininity or womanhood, and low pitch to be a sign of masculinity or manhood. Associations like these are known in sociolinguistics as 'indexical', as the relationship is imagined to be causal or co-occurrent, much like the indexical link between smoke and fire.

It bears emphasis that the process of interpreting variations in language use (such as variations in voice pitch) as meaningful signs depends on perceivers' understanding of the world (Gal and Irvine 2019). Some indexicalities, such as the indexical link between voice pitch and gender, are the products of understandings of the world which align with and support structures of power (Calder 2021). These indexicalities come to be entrenched as expectations and norms, benefiting those who fit into gendered ideals and marginalizing those who do not. Although many cultures may have somewhat overlapping gender ideals, including the cross-culturally common idealization of the feminine voice as higher-pitched and the masculine voice as lower-pitched, the character of these indexicalities and their applications is nonetheless culturally specific. In the western, English-speaking, North American cultural context of this study, gendered indexicalities such as this one are embedded in an understanding of gender which idealizes those in power and privilege here: white, cisgender, heterosexual, middle-class men and women. These norms serve to marginalize those outside of this group, deeming their language use inadequate and inappropriate (eg. Inoue 2006; Inoue 2003; Flores and Rosa 2015; Heffernan 2023).

In investigating indexicality, existing sociolinguistic studies primarily focus on only one set of these expectations, either the normative, powerful set described above (eg. Hazenberg 2012; Podesva and Van Hofwegen 2015) and henceforth referred to as 'hegemonic', or alternative, local sets constructed within marginalized communities (eg. Mendoza 2024; Calder 2019a). Within the realm of the study of trans speech, the former type of study often interprets trans voices in their deviance from or conformance with cisgender speech, and runs the risk of tacitly enforcing hegemonic norms, marginalizing the diversity of trans voices (Calder and A. Steele 2025). Furthermore, focus within a single interpretive lens – hegemonic or otherwise – often sidelines the intersubjective nature of identity, by which identity emerges in the interaction of subjects and their establishment of contrast between self and other, speaker and listener (Bucholtz and Hall 2010). Marginalized speakers are compelled to hold awareness both of hegemonic and alternative indexicalities, and to predict how their listener will interpret their speech.

Recent research on transgender speakers has worked to take up this perspective, placing more even focus on both sides of the speaker-listener pair by investigating how trans voices are perceived by listeners (Jessee and Calder 2025), and trans people's understandings of these perceptions (Heffernan 2023). However, prior research has not investigated how these understandings manifest in trans speak-

ers' actual linguistic practice. A mainstay of sociolinguistics is the study of changes within individual speakers' language use in response to changes in context, known as style-shifting. Many style-shifting studies investigate individuals' speech across conversations with different people, finding changes in response to interlocutor(s) – who they are talking to (e.g. Podesva 2007; Rickford and Mcnair-Knox 1994). This body of research on interlocutor-oriented style-shifting, however, has not investigated trans speakers' style-shifting in response to the trans status of their interlocutor – whether the person they are talking to is cisgender or transgender. By filling this gap, I provide data on how transfeminine speakers navigate cisgender linguistic norms.

Much of the existing study on interlocutor-oriented style-shifting primarily finds that speakers often change their linguistic behavior to be more similar to the speech of their interlocutor, known as accommodation, or audience design (eg. Rickford and Mcnair-Knox 1994). This practice is, to some extent, a sociolinguistic fact of life – all speakers accommodate sometimes and to some degree. It is this universality, however, which precludes analyses which center accommodation from meaningful interpretations of the social motivations behind style-shifting (see eg. Podesva 2007; Rickford and Mcnair-Knox 1994). My study, which aims to understand the influence of cisgender norms on transgender speakers, does not center accommodation, instead considering speakers' style-shifts in their implications for speakers' understanding of and navigation of hegemonic norms (see section 4.1 for more on accommodation).

Language and gender research has found that, similar to voice pitch, the English /s/ sound varies in pronunciation, and that hegemonic gender norms expect women to pronounce /s/ with a fronter tongue position (closer to the teeth) than men (Flipsen et al. 1999, Hazenberg 2012; Zimman 2013; Podesva and Van Hofwegen 2015). I focus on this sound for two primary reasons. First, unlike voice pitch, gendered variation in /s/ is not influenced by gendered anatomy; voice pitch is influenced by the size of the vocal tract, whereas /s/ pronunciation is not (Zimman 2017c). Second, research has found that when men pronounce /s/ with a fronter tongue position than norms expect them to, their voices are perceived as sounding gay (e.g. Calder 2019a). For transfeminine speakers, this makes /s/ particularly difficult territory, as it is possible that moving toward fronter, hegemonically feminine /s/ pronunciations may only result in being misperceived as gay men. In order to study how trans speakers navigate hegemonic linguistic norms, I look to this difficult situation, investigating how transfeminine speakers shift /s/ pronunciations from a conversation with a transgender listener to one with a cisgender listener, and what factors may influence the (conscious or unconscious) decision to shift.

There are three possible outcome behaviors: speakers shift towards fronter, more hegemonically feminine /s/ pronunciations with the cis listener, speakers do not shift, or speakers shift towards backer, less hegemonically feminine /s/ pronunciations with the cis listener. I initially predicted the first outcome would be most likely from speakers who are unconcerned with being mistaken for gay men but who do wish to avoid being judged as insufficiently feminine by cis listeners.

The other two outcomes (either no shift, or a shift towards backer /s/ pronunciations with the cisgender listener), in my initial predictions, are most likely for speakers who differ from this description: those who are concerned with being mistaken for gay men, and those who are unconcerned with cisgender perceptions or hegemonic conceptions of femininity.

In order to test these predictions, I recruited 4 transfeminine speakers and 2 women conversation partners, one trans and one cis. I conducted 20-30 minute one-on-one audio recorded casual conversation sessions between each speaker and each conversation partner, in which participants were asked to collaboratively play a word game and then to chat casually and get to know each other. I then acoustically measured their /s/ pronunciations in each recording. After the conversation sessions, I interviewed each speaker, asking questions about their relationship with gender, sexuality, and their voice, with a focus on the challenges of dealing with cisgender perceptions, and assessed the extent to which speakers expressed the concerns pertaining to my predictions.

I find that two speakers shift towards fronter, more hegemonically feminine /s/ pronunciations with the cisgender listener, one speaker does not shift, and one speaker shifts away from hegemonically feminine /s/ pronunciations with the cisgender listener. Counter to my predictions, the former two speakers (who shifted towards more hegemonically feminine /s/ pronunciations) were the two who expressed concern about sounding like gay men, and the latter two speakers did not express these concerns. Also counter to my predictions, all speakers seemed similarly concerned with cisgender judgments, and the speaker who shifted to backer /s/ (away from the hegemonically feminine pronunciation) was the only one who did not strongly distance herself from hegemonic femininity.

In interpreting these results, I first return to my initial question: whether speakers' shifts can be interpreted as reacting to the trans status of their conversation partner. In order to answer this question, I consider whether the behavior of the two speakers who shift (towards hegemonically-feminine /s/ pronunciations) can simply be explained as accommodation to the way their conversation partner pronounces /s/. Although I am unable to conclude with certainty that these speakers are not simply accommodating, I nonetheless consider it more likely that they are responding to the trans status of their conversation partner for two reasons. First, my speakers emphasize the influence cisgender perceptions have on their relationships with their voices in my interviews with them, encouraging an interpretation which considers this influence. Second, the accommodation approach does not provide sufficient explanation as to why two speakers shifted /s/ in the predicted direction, and two did not.

In answering this question, I return to my initial predictive factors. I notice that the the two pairs of speakers are indeed differentiated by their expression of concerns with sounding gay-masculine, but not in the way I predicted. I initially predicted that these concerns lead speakers to not shift, in order to avoid indexing gay-masculinity. Instead, it seems that these concerns would lead speakers to be more (consciously or unconsciously) aware that /s/ is an important indexical resource (than those who do not voice these concerns). Under this interpretation,

this awareness spans both indexical meanings of /s/ – gay-masculinity and femininity – and speakers take up the feminine meaning to protect themselves from cisgender listening judgments (in the hopes that the gay-masculine meaning will not be perceived as relevant to them). With respect to my other predictive factors of cisgender perceptions and relationship with hegemonic femininity, I find that speakers primarily seem to prioritize their safety, rather than disregarding cisgender judgment. I also discuss other possible factors motivating shifts, including transition stage and race.

From these interpretations, I conclude that trans speakers' awareness of hegemonic cisgender speech norms does indeed influence their linguistic behavior, but that this influence is mediated by a sufficient (conscious or unconscious) awareness of a given form's indexicality. I conclude that speakers make use of complex understandings of indexicality to predict how they will be perceived, and change their speech to navigate these perceptions. From these conclusions, I emphasize the difficulty of trans speakers' linguistic position, the injustice of hegemonic norms, and the marginalizing influence of privileged listening practices. I highlight the power speakers have to protect themselves from these norms, and acknowledge the injustice that this self-defense is necessary.

Chapter 1

Literature Review

1.1 Theoretical Motivations and Background

This thesis belongs to a lineage of research in the field of language and gender. Scholars of language and gender look to the gendered indexicalities of everyday linguistic behavior as a medium for gender performativity. For the bulk of this thesis, I follow the tendency of language and gender scholarship in centering indexicality as the primary framework of analysis. Although performativity is the point of departure for much of language and gender research – especially following the importance of performativity within gender studies after the foundational work of Judith Butler (1990) (see also Bucholtz 2014) – indexicality provides a number of theoretical tools, as reviewed below, which make for a more actionable lens of analysis (Henry and Hall 2025). Sociolinguists study the "social-indexical performativity of language-in-use" (Silverstein 2023, p. 17), considering performativity to be mediated by indexicality (Henry and Hall 2025). For this reason, I move forward by investigating indexicality. In this chapter, I first introduce recent approaches to indexicality in language and gender research, demonstrating the turn towards an approach which considers speaker and listener together, and underlining theoretical questions which still remain. I then review the language and gender literature on /s/, arguing that the existing scholarship is unable to provide complete answers to these theoretical questions.

Indexicality comes to sociolinguistics through semiotics, the study of meaning-making. In this thesis I adopt the model of semiotics put forth by Gal and Irvine (2019), who synthesize prior approaches, placing critical emphasis on the procedural aspect of meaning-making by which phenomena come to be meaningful. In their model, individuals make meaning of their world through conjectures. Drawing upon their knowledge and understanding of the world, they recognize a phenomenon and conjecture that it may be a sign, i.e. it may be meaningful. They conjecture that the sign signals (means or represents) a particular object, its meaning. Essential to note here is that these conjectures constitute the sign-object relationship; nothing is a sign in itself, independent of a perceiver conjecturing that it is a sign which signals some object (Gal and Irvine 2019, p. 89). Also important is

that these conjectures are based on the perceiver's knowledge and understanding of the world, and which are constructed and mediated by their cultures, communities, and individual social identities– their lived experience (Gal and Irvine 2019; see also Calder 2021). Part of a conjecture is its "grounds": the reasoning the conjecture has as to why its phenomenon is a sign and why it signals this object. A conjecture is considered to be 'indexical' if its grounds are the knowledge or belief that the sign often co-occurs with, points to, or is caused by its object (Gal and Irvine 2019, p. 96).

Sociolinguistics often focuses on finding correlations between differences in language use (eg. pronunciation, syntax, word choice) and differences in social identity, persona, or context (Eckert 2012). Particular linguistic forms are then said to index the social identities of those who use them, on the grounds of this relationship of co-occurrence (Eckert 2012). As discussed using the example of voice pitch above, if a speaker uses a particular linguistic variant (e.g. high-pitch), and the listener believes that form to be commonly used by those of a particular gender (e.g. women), they may make an indexical conjecture on the grounds of this co-occurrence, considering the use of that form to index that speaker's gender. That there are two actors in play here, a speaker and a listener, led Bucholtz and Hall (2005; 2004; 2010) to understand the indexical-performative construction of gender (and identity categories more broadly) as an intersubjective process – a process which takes place through interaction of two or more people.

Bucholtz and Hall characterize this intersubjective process as a negotiation between the indexical goals of the speaker and the indexical goals of the listener (eg. Bucholtz and Hall 2005, p. 606). Although it is possible that a listener may construe an indexical meaning from a linguistic form used by a speaker who has no knowledge of, or intent to make use of, that form's indexicality, agentivity or intentionality on the part of the speaker is common (whether conscious or unconscious). Speakers have interactional goals, and hope that their behavior will be construed as indexical of those goals, whether it be related to identity, eg. presenting feminine, or other aspects such as persona and stance, eg. seeming like a 'diva' (Podesva 2007). This agentive view arose principally in reaction to earlier sociolinguistic models which stripped speakers of agency and individuality, instead understanding their linguistic behavior simply as a product of their static, essentialized identities (Eckert 2012). In practice, this turn reframes "she speaks this way because she is a woman" as "she chooses this form to index femininity." Acknowledging that individuals have choices about how to present themselves and granting them authority over their self-identification is also a move towards alignment with queer and trans understandings of identity; trans people in the US often consider gender identity to be self-determined (Zimman 2017b; Heffernan 2023), rather than a consequence of bodily characteristics. Despite the value of emphasizing the speaker's goals to legitimize their identities and agency, it is nonetheless the case that the listener is the powerful other side of this negotiation.

Inoue (2006; 2003) argues that in cases where the listener occupies a privileged social position of power over the speaker, they may inhabit a privileged 'listening subject' position in which the beliefs required to uphold this social hierarchy may

play a primary role in their interpretation of indexical meanings, overriding or erasing the speaker's intentions. Studying the emergence of the notion of 'women's language' in Japan, Inoue finds that to Meiji period elite male intellectuals, schoolgirls inhabited a social position which represented a modernizing breach from traditional Japanese social structure, and a threat to their power. For this reason, when they overheard schoolgirls using the verb ending forms *teyo* and *dawa*, they construed these endings as indexical of degeneracy and social decline. Of course, schoolgirls themselves did not intend to use these forms to index a 'degenerate' identity; the purported degeneracy these forms were said to represent was a social fabrication rather than a truth about any actions these girls were involved with. In this way, male intellectual overhearers constructed a listening subject position which made negative judgmental conjectures about the indexicality of schoolgirls' speech, and constructed a fictional object for the these speech forms to signal. The male listening subject sets the narrative, erasing schoolgirl speakers' semiotic agency.

Following this important insight, research has extended theorization of the listening subject to consider race and queerness, researching white (Flores and Rosa 2015) and cisgender listening subjects (Calder 2021; Heffernan 2023; Jessee and Calder 2025; Calder and A. Steele 2025). Flores and Rosa (2015) find that when white listeners racialize speakers as non-white, they justify their position of power over those speakers by construing their speech as indexing an underspecified 'inappropriateness' for the situation (eg., the classroom), regardless of those speakers' adherence to linguistic norms for standard English, constructing a white-supremacist white listening subject which will always deem non-white speech deficient and indexical of a supposed non-white inferiority. Jessee and Calder (2025) find that cisgender listeners apply stricter gendered listening standards to an audio clip of a voice when they are told the speaker of the clip is a trans woman than when told the speaker is a cis woman, similarly situating sufficient gender performance out of reach for transgender speakers in a move to deny their identities.

Drawing on these concepts of white, cis, and male listening subjects, Calder (2021) and Calder and A. Steele (2025) raise an important critique of the sociolinguistic project, questioning which listening subjects researchers themselves may be inhabiting in their interpretation of speakers' language use. Following the conjecturing model of Gal and Irvine (2019), they note that researchers collect numerical correlation data between linguistic forms and social aspects, and use this as the knowledge on which to base their conjectures of indexical links. Incorporating the research on the listening subject, they conclude that although this data may seem objective, empirical, and scientific, purely describing the world as it is (see also Zimman 2021a), listening practices inevitably are biased by the sociocultural positioning of the researcher and the scholarship that researcher builds upon. This critique raises the question of how to research the social-indexical performativity of language-in-use without taking up and reinforcing violent listening practices and marginalizing speakers who are not white, cis, heterosexual, middle class men. Work which addresses this question employs advanced models of indexicality which encourage sociolinguistic study to move from simply asking what

meanings participants index to investigating the socio-cultural structure of individuals' indexical knowledge, and the intersubjective process of indexicality (Calder 2021; Silverstein 2003; Silverstein 2023; Barrett and Hall 2024). The concepts of the indexical field as adapted by Calder (2021), and indexical (mis)recognition as developed by Barrett and Hall (2024) have been used to model indexical knowledge and the relativity and intersubjectivity of indexicality.

The concept of the indexical field, introduced by Eckert (2008), has been adopted by Calder (2021) to capture the way in which a particular phenomenon may be conjectured to have a multiple meanings, both within a single group of people and across different groups of people. The concept of the indexical field emphasizes, connecting to Gal and Irvine's (2019) semiotic model, that indexical relationships are conjectured through listeners' knowledge. In this way, a particular linguistic form may be construed as a sign of any number of meanings, depending on which knowledge the listener uses to conjecture the relationship, and what conclusion they come to in conjecturing. Although this knowledge is mediated and constructed differently by each individual's social and cultural lived experience, it is nonetheless to a large extent shared. For example, Inoue's (2003) listening subject was not a single Meiji-era Japanese male intellectual, it was an ideological position constructed and inhabited by this male intellectual social class. Individuals who inhabit similar social and cultural backgrounds and social positions, as well as co-members of communities, are likely to draw upon a similar base of knowledge, and are likely to consider certain phenomena signs with a particular set of possible meanings. Calder (2021) conceptualizes the indexical field as this set of meanings which may be shared by a set of people with shared epistemology. By making explicit whose indexical field researchers are investigating and interpreting signs through, the fact that this interpretation is coming from particular perspectives is forefronted, preventing the universalization of the hegemonic indexical field.

While this makes progress towards a more critical and less biased move to interpreting signs, and much research has been productively done to investigate the indexical fields within particular local communities, especially community-specific systems of categorization and meaning, it is nonetheless untenable to simply split the world into non-overlapping spheres of shared knowledge and shared indexicality. Individuals bring their own indexical knowledge into interaction, and are likely aware of a number of indexical fields. The question that arises is: what happens when listeners apply an indexical field differently than a speaker intends them to, or when there is a mismatch in indexical knowledge?

Barrett and Hall (2024) model these situations using their concepts of the indexical inventory and indexical (mis)alignment/(mis)recognition. For them, an individual's indexical knowledge is structured in an indexical inventory, which is "an individual's full set of indexical fields." (p. 132) When individuals interact and engage in sociolinguistic meaning-making through shared indexical fields, they are said to be in indexical alignment. In this case, if a speaker uses a linguistic form that they know to hold a particular indexical meaning within one of the indexical fields in their inventory, their listener would recognize (in Barrett and Hall's term)

their behavior as indexical of the meaning on which they are aligned, interpreting it in line with the speaker's intentions. However, in cases of indexical misalignment, the speaker's intended interpretation does not come through: the speaker employs a linguistic form with the intent to index a particular meaning in their inventory, but the listener does not interpret the form as indexing that meaning. Although it is possible the listener may simply be unaware of the indexicality the speaker intended to make use of, Barrett and Hall (2024) emphasize that misrecognition is often politically motivated, whether this motivation be resistance against, or reproduction of, hegemonic power.

For the purposes of this thesis, it is most relevant to review the marginalizing side of misrecognition, by which misinterpretation of signs is a tool for the reproduction of power. Barrett and Hall (2024) raise the example of the transition of generic language in English from masculine (eg. *he/mankind*) forms to gender neutral forms (eg. *they/humankind*). They note that in this transition, feminists constructed a new indexical meaning around generic masculines, considering these forms an index of misogynistic views in those who use them. In reaction, anti-feminists deny this new indexicality, considering generic masculines grammatical fact and misrecognizing, or refusing to recognize, new meanings. Here, Barrett and Hall consider the anti-feminist approach to be an example of their concepts of indexical presumption – whereby those in power presume that their indexical inventory is universal and correct – and indexical denial – in which knowledge of indexicalities against one's interests are denied.

When considered together with the concept of listening subjects, indexical presumption and indexical denial emerge as tactics which privileged listening subject positions might employ in interpreting meanings from linguistic behavior. Taken as a whole, the concepts of the indexical field, listening subjects, and tactics of indexical misrecognition do the work of, in Barrett and Hall's terms, relativizing indexicality. This makes it no longer tenable for researchers of sociolinguistic indexicality to draw correlations and make their interpretations from there; instead, they must work to understand the procedure by which meaning arises, and all of the factors which influence speakers' choices in language use. In particular, this thesis is interested in asking how marginalized speakers deal with these listening practices, hoping to contribute to an understanding of how speakers model listeners' indexical inventories and how they make use of these models in making (conscious or unconscious) linguistic choices.

Although all speakers may to some extent model their listeners' indexical inventories, I choose to focus on the case of trans speakers as a marginalized group under particular pressure to do so. Barrett and Hall (2024) note that indexical meanings are not evenly distributed, and I would add that the work of modeling listeners' indexical inventories is also not evenly distributed. When marginalized speakers interact outside of safe communities, they are at risk of being judged negatively by listening subjects which will presume the hegemonic indexical field and deny alternate indexicalities. Conversely, privileged speakers are unlikely to face any negative consequences due to ignorance of indexical fields belonging to marginalized groups; marginalized listeners do not have the power to inflict the

same violence as privileged listeners might. In reaction to this situation of imbalance, marginalized speakers are coerced to protect themselves by learning hegemonic indexicalities and predicting how listeners may interpret their speech from an antagonistic listening subject position.

Heffernan (2023) reveals that transfeminine speakers are aware of the cisgender listening subject. She collects metalinguistic commentaries on transfeminine individuals' relationships with their voices and experiences with being heard from two sources: an audio-recorded conversation between two US (South Asian American and White American) trans women, and posts made by anonymous transfeminine users on the Reddit online forum r/TransVoice. Commentaries note that cisgender listeners hear transfeminine voices through hegemonic norms and make binary judgments, categorizing voices as belonging to a man or a woman. Transfeminine speakers do not want to sound like men, and are aware of the possibility that cisgender listeners may make this judgment. In these commentaries, speakers additionally discuss what they do with this awareness and how they respond to this risk, a question I take as central to my study.

Heffernan finds that speakers simultaneously express feelings of pressure to modify their voices to avoid these negative judgments, and feelings that they should not have to do so, exposing the conflict between "views of gender as self-determined, on the one hand, and of gender as intersubjectively constituted, on the other" (ii). These two approaches can be found in the commentaries from one of the audio-recording participants. This participant notes that when "interacting with the cis universe" she finds herself "deliberately modulating [her] voice" to make it "more girly," and that this is "a kind of survival tactic... to try and avoid any possibility that [she]'ll get misgendered" (23). In this way, the speaker acknowledges the risk of misgendering that comes from interacting with cisgender listeners, and responds by changing her language use in an attempt to align more closely with hegemonic feminine norms. On the other hand, she also asserts that "it's kind of just always incorrect to assume someone's gender," and that she is uncertain about whether she would want to train herself to always use a more hegemonically feminine voice, as she wants to be able to "[think] about [her own] voice ... through uh- like the lens of other trans people" (30), who would not apply the hegemonic indexical field, not assume her gender, and who would accept her voice as it is. This tension around the decision to shift one's speech in the presence of a privileged listener who may inhabit a judgmental listening subject position is of primary interest to my study. Because she only investigated metalinguistic commentaries, Heffernan was not able to investigate how these decisions play out in transfeminine people's real interactions with cisgender listeners; this thesis aims to fill this gap.

Within the existing literature, I identify three style-shifting studies which include transgender speakers: Becker et al. (2022), Gratton (2016), and Parnell-Mooney (2019). Becker et al. (2022) find that trans men and nonbinary people on testosterone (assigned female at birth) use significantly less creaky voice (known popularly as 'vocal fry') in a reading-passage task compared to in a open-ended conversation. As creaky voice is associated with femininity within the North American

English hegemonic indexical field, they tentatively interpret these findings as evidence that in a more scrutinized form of speech (in which, I add, the notion of the cisgender listening subject may be more mentally present), these individuals purposefully move away from normative feminine speech in order to assert their identities as non-women.

This aligns with the findings of a study on a Toronto nonbinary friend group conducted by Gratton (2016). Gratton finds that alternation between *-ing* and *in'* (eg. *working* vs. *workin'*) was more even in queer-safe spaces, where speakers used roughly the the same amount of both, and more differentiated in public, where speakers preferred one form over the other. In the hegemonic indexical field, the *-ing* variant is hegemonically associated with femininity, whereas *-in'* is hegemonically associated with masculinity. Gratton finds that in public, speakers avoid the variant which hegemonically indexes their assigned gender at birth, understanding that cisgender overhearers are invested in imposing that gender onto them – denying their nonbinary identities – and choosing to exaggerate their difference from that gender to avoid that perception. In queer-safe spaces, (in which, I add, the cisgender listening subject is not present), speakers were free to speak more neutrally and without regard for binary norms – Gratton notes that one speaker assigned male at birth mentions feeling comfortable making occasional use of a 'manly voice' in these spaces, as they know they will not face misgendering for doing so in the absence of cisgender listeners.

Parnell-Mooney (2019) also finds this sort of differentiation, finding that in conversations with cis listeners, four trans men and two nonbinary speakers shift to pronounce /s/ further from the type of pronunciation expected of their assigned gender at birth. The trans men and the nonbinary person assigned female at birth shift further from hegemonically feminine /s/ pronunciations with cisgender women listeners than when interacting with cisgender men, whereas the nonbinary person assigned male at birth shifts the opposite way. Although these studies present important empirical results, they do not integrate with the theoretical material discussed above – listening subjects, indexical fields, indexical recognition, and the modeling of a listener's indexical knowledge. They also do not include transfeminine speakers, and do not directly compare speech with a cis interlocutor against speech with a trans interlocutor. Filling these gaps, I compare transfeminine speakers' indexical behavior across a conversation with a cisgender woman and a conversation with a transgender woman, interpreting shifts in light of speakers relationships with their gender, their voice, and being heard as a transfeminine speaker.

For my comparison, I focus specifically on the English /s/ sound, due to the richness of existing research on its indexicalities and its use by queer and trans speakers, especially in connection to the sophisticated theories of indexicality discussed above. The remainder of this section introduces the literature on /s/ and reviews its connections to the concerns of transgender language and indexicality research.

1.2 The Sociophonetics of /s/

By varying the tongue position with which an /s/ is pronounced, a speaker can vary the frequencies (or pitches) of the sound produced; when the tongue position is fronter (closer to the teeth) the resulting /s/ has primarily higher frequencies (sounding like higher pitch), when the tongue position is backer (further from the teeth), the resulting /s/ has primarily lower frequencies (sounding like lower pitch) (Fuchs and Toda 2010). A common measure of this frontness is the spectral center of gravity (CoG), which is the mean of all of the frequencies of an /s/ sound at a given moment in time, weighted for their amplitudes (volumes), measured in Hz. It is important to note at the outset that gendered differences in /s/ frontness do not arise from any anatomical differences associated with the sex binary, unlike other gender-correlated sites of variation such as overall vocal pitch; all people with typical vocal tracts have the anatomical equipment to produce roughly the same range of /s/ pronunciations (Zimman 2017c; 2021b).

Research on white, English-speaking, cisgender heterosexual men and women in North America – which did not include non-White, non-English-speaking, or queer speakers and listeners, a point to which I will return – finds robust correlations between speaker gender and /s/ pronunciation. Flipsen et al. (1999) review 21 such studies, finding that women's /s/ pronunciations are consistently fronter than men's. Listeners in this population are aware of these correlations, and judge fronter /s/ to sound feminine, and backer /s/ to sound masculine. Strand (1999) finds that when asked to categorize audio stimuli on a spectrum of /s/ frontness as either /s/ or /ʃ/[1] (/ʃ/ is acoustically similar to a very backed /s/), they place the boundary higher when told the speaker of the audio clip is a woman than when they are told it is a man, thus exhibiting a link between /s/ frontness and perceived gender. When listeners in this population perceive a mismatch in /s/ pronunciation and the perceived gender of a voice, they judge that voice to sound gay or lesbian. A number of studies have focused on this dynamic in perception of men's voices (eg. Levon 2014; Campbell-Kibler 2011). Munson, Jefferson, and McDonald (2006), replicate Strand's experiment, finding when listeners perceive a speaker to be lesbian sounding, they place the boundary between /s/ or /ʃ/ lower, thus expecting lesbians to back /s/.

Calder (2021) points out that these earlier correlational studies often interpret indexicalities, but fail to situate these indexicalities as specific to the populations in which they are found. It is insufficient to claim that fronter /s/ pronunciations index femininity and gayness and backer /s/ pronunciations index masculinity and lesbianness without specifying for whom these indexicalities hold. These indexical links are made specifically by cisgender heterosexual white listeners, who take the behavior of their own population as the norm, and interpret others through this norm (an act of indexical presumption). Calder (2021) thus considers indexical links from /s/ frontness to femininity and gayness to be gender norms within the hegemonic indexical field for /s/, rather than neutral and universalizable patterns.

[1] /ʃ/ represents the English 'sh' sound, as in the beginning of 'shirt' or the middle of 'station'

Calder (2021) proceeds to discuss two examples of research outside of this hegemonic indexical field, one investigating Black American speakers' /s/ pronunciations, and one investigating queer nonbinary drag queens'. In the first of these studies, J. Calder and King (2022) find that for cisgender heterosexual Black speakers in Bakersfield, California, /s/ pronunciations were not differentiated by speaker's gender. They highlight that white normativity does not apply here; the men and women who participated in this study considered themselves normatively gendered cisgender heterosexuals, and cannot be said to be improperly indexing their genders, or indexing queerness. Within the indexical field of speakers in this community, Calder and King argue /s/ comes to be an index of white femininity/masculinity, and thus irrelevant to (or even to be avoided by) Black speakers.

Looking at racialized aspects of /s/ in perception, Maegaard and Pharao (2015) and Pharao et al. (2014) present a similar finding in a different context. In white cisgender heterosexual Copenhagen Dutch, /s/ has been shown to pattern and be perceived similarly to the North American white cisgender heterosexual English-speaking population. Maegaard and Pharao find that teenage listeners (both white and nonwhite) only hear fronted /s/ as feminine and gay when it is within the context of a prosodic pattern associated with middle-class whiteness – 'modern' standard Copenhagen Danish speech; in the context of a prosodic pattern associated with lower-class and immigrants – 'street' Danish – listeners instead hear fronted /s/ as associated with 'gangster'-ness and masculinity. Although Maegaard and Pharao do not apply the concept of indexical fields to their work, it can be said that /s/ has two indexical potentials within the indexical field of these listeners, and that other indexical behavior, in this case the prosodic pattern, determines which set of indexicalities speakers take up – if the speaker sounds white with the 'modern' prosodic pattern, /s/ is interpreted through white gender norms, if the speaker sounds non-white with the 'street' prosodic pattern, /s/ is interpreted differently.

From these examples, two points are clear. First, local or marginalized communities' indexical fields are not simply independent of hegemonic meanings, but are instead interlinked with them, qualifying and reinterpreting hegemonic meanings. Second, variables like /s/ do not work on their own, but only come to be interpretable through a broader semiotic context; listeners in Maegaard and Pharao's studies could only interpret /s/ through its combination with prosodic types. In the second of the studies mentioned above on /s/ within a community of queer drag queens, Calder (2019a) applies the theories of indexical orders developed by Silverstein (2023; 2003), and fractal recursivity developed by Irvine and Gal (2000) and Gal and Irvine (1995) in order to investigate this process.

As linguistic forms are construed as indexical signs and gain meaning within a particular indexical field, it is possible that new meanings will arise which build off of and are informed by the original meaning. These meanings form an ordered structure, in which some meanings are taken to be more basic and said to be at a lower indexical order, and some meanings are taken to be more derived, and said to be at a higher indexical order. The development of an indexical link between a linguistic form and a social meaning always already emerges informed by

an uncountable chain of previous indexicalities and meaning-making, and for this reason, the first or lowest relevant indexical order to a given study is named the nth indexical order, and subsequent orders are expressed as the $n+1$st, $n+1+1$nd, and so on (Silverstein 2003). Development of higher indexical orders can take place both within a single indexical field, and cross-cutting the overlaps of multiple. In the case of /s/, Calder understands the indexical link between the gender binary and /s/ frontness to exist at the nth order within the hegemonic indexical field. When cisgender heterosexual white listeners' perception of speakers' gender does not match with their perception of that speaker's /s/ pronunciation, this recontextualization of /s/ usage forms the $n+1$st indexical order, in which a new interpretation must emerge. Here, the meaning of /s/ comes to be qualified by other construals of meaning, in that a listener's perception of a speaker's gender must be based upon other indexical conjecturing. That cisgender heterosexual white listeners construe this perceived mismatch in speaker gender and /s/ pronunciation as gay/lesbianness evidences an erasure of transgender, nonbinary, and other queer identities; gender is assumed to be fixed, and all gender nonconformance is assumed to be gay or lesbian. Without these cisgender assumptions, the $n+1$st indexical meaning of a mismatch of perceived speaker gender and /s/ would simply be gender nonconformance, or perhaps that the speaker's gender was perceived incorrectly. With the prior assumptions, it comes to be gay/lesbianness. It is important to note that although the nth and $n+1$st indexicalities of /s/ are situated at different orders, they nonetheless inform and construct each other, and do not form a chronological order despite their ordered structure.

Within the San Francisco drag communities Calder (2019a) studies, an additional indexical order of /s/ is forged at the $n+1+1$nd level which is local to these communities' indexical fields, escaping the hegemonic field. A drag community Calder studies in the South of Market (SoMa) neighborhood of San Fransisco figures itself as more radically queer than another drag community they study in the Castro neighborhood. This is reflected by the differences in identities between these communities; SoMa queens generally identify as beyond the gender binary, many of them nonbinary or transgender, whereas the queens of the Castro neighborhood primarily identify as cisgender gay men. Calder finds that SoMa queens front /s/ more than Castro queens, and that this forges a new local indexicality to 'fierceness', anti-normativity, and in-group status at the at the $n+1+1$nd level. Their diagram of this structure of meanings is included in figure 1.1. This emerges as an example of indexical orders cross-cutting indexical fields, where meanings in the hegemonic cisgender heterosexual white indexical field came to inform the creation of new meanings in the local, alternative indexical field.

As mentioned above, within Calder's model fronting /s/ emerges as deviant from gender norms only in those whose gender is understood as 'male' within the cisnormative conception of gender as bodily characteristics. It is perhaps surprising, then, that nonbinary and transfeminine speakers in the SoMa drag community, who would not like to be perceived as men, front /s/ exactly for the purpose of gender defiance, rather than gender conformance. It is helpful here to introduce a theoretical tool Calder employs to understand the interlinking of identity

categories at these different levels of sex at birth, gender identity, sexuality, and trans/queerness– the semiotic process of fractal recursivity, as theorized by Irvine and Gal (2000) and Gal and Irvine (1995).

In fractal recursitivity, a semiotic opposition on one level becomes nested into itself, and this process can happen any number of times (Irvine and Gal 2000; Gal and Irvine 1995). In moving from the nth indexical order of /s/ to the $n+1$st (both of which are within the hegemonic indexical field), the gender binary is nested into itself, creating a metaphorical structure where men and women are split into binary gender oppositions. This can be understood as distinguishing some notion of 'men who are men' from 'men who are women', and 'women who are women' from 'women who are men'. These become the metaphorical representation of gay men and lesbian women, since members of these groups are still fundamentally understood as types of men and types of women, relying on the aforementioned erasure of trans people and those beyond the binary. Is it then the case that by situating the distinction between cisgender gay Castro queens and nonbinary and transgender queer SoMa queens as nesting into the 'men who are women' branch, Calder as the analyst, and possibly the SoMa queens themselves, tacitly consider these queens as nonetheless a 'type of man/male'? I argue it is not. Instead, in my reading of Calder's work, SoMa queens reinterpret at the $n+1+1$st level such that the fractal-recursive binary structure is broken; instead, the opposition is cisgender gay men vs. non-cisgender queer people, or as Calder articulates it, gay vs. queer. This binary still fits with the base-level nth order of /s/ indexicality, which is always present; it is not difficult for SoMa queens to figure gay men as less feminine and less subversive than queer non-cisgender people. From here, the indexicality of /s/ frontness as subversive queer non-cis femininity can be less problematically interpreted.

A. J. Steele (2022) provides another example of /s/ being reinterpreted by those outside of the gender binary. They investigate a racially-diverse community of Columbus, Ohio nonbinary individuals, finding that for Black speakers, fronted /s/ seemed to have the opposite indexical meaning, correlating positively with speakers' self-rating of their masculinity. For white speakers in this community, there was no correlation between /s/ pronunciations and masculinity ratings. This leads Steele to conclude that Black speakers subvert /s/ meanings not only to create a nonbinary identity and differentiate themselves from a cisgender norm, but also to repurpose and subvert a white norm. For white speakers, this dimension is not present, so the behavior is different. These patterns suggest that nonbinary speakers in this community forge further indexical orders which allow them to negotiate their specific racial and gender positions against white and cisgender normativities.

The studies discussed above illustrate the ability of speakers outside of cisgender heterosexual white populations to construct alternative and subversive meaning structures for /s/ which allow an interpretation of their unique speech practices as innovative and subversive, rather than inadequate and deviant. However, due to the dominance of the hegemonic indexical field, and the practicalities of social life, it is nonetheless the case that marginalized speakers outside of cisgen-

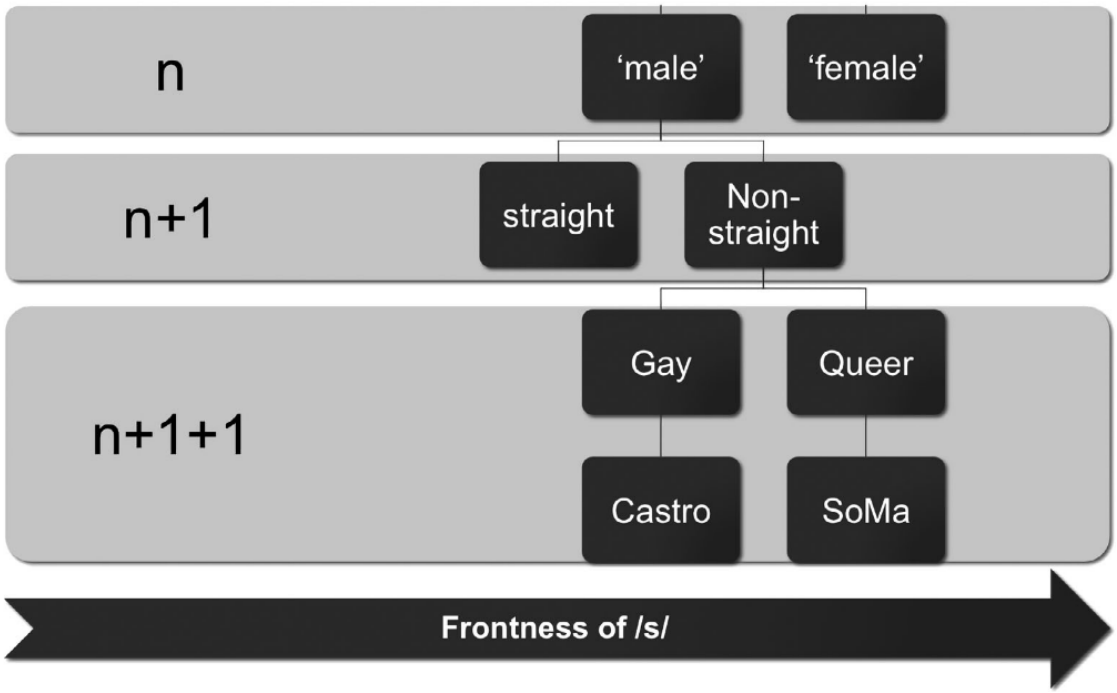


Figure 1.1: Diagram from Calder (2019a, p. 340) illustrating indexical orders of /s/ frontness and corresponding fractal recursive social oppositions across two San Fransisco drag communities.

der heterosexual white populations have to deal with these norms. In addition to the fact that that all speakers are socialized to believe in these norms, and may apply norms which marginalize themselves, marginalized speakers inevitably interact with privileged listeners, who are most likely to apply these norms. In this light, I ask how marginalized speakers deal with the hegemonic structure of meanings when forming new indexical orders or rejecting hegemonic meanings may be risky: interactions with privileged speakers who are likely to apply hegemonic interpretations.

That the hegemonic structure of indexicality for /s/ particularly excludes trans speakers is attested to by the findings of Jessee and Calder (2025). They investigate cisgender perceptions of transgender /s/ by replicating Strand (1999), comparing the placement of the boundary between /s/ and /ʃ/ when the perceiver is told the speaker is trans vs. not told they are trans (and thus is likely to assume that they are cis). They find that unlike in Munson, Jefferson, and McDonald (2006), where cisgender heterosexual speakers expect cisgender gay men and lesbians not to conform with the hegemonically normative /s/ pronunciation for their gender, transgender speakers are expected to hyper-conform– trans women are expected to front /s/ more than cis women by all listeners, and listeners who do not know a trans person also expect trans men to back /s/ more than cis men. Although indexicalities certainly would have changed in the 20 years between these two studies, it is nonetheless surprising that cisgender heterosexual white listeners hold dif-

ferent expectations and perceptions for trans people than for cisgender gay men and lesbians, especially since these expectations run counter to trans speakers' actual behavior; Podesva and Van Hofwegen (2015), Hazenberg (2012), and Zimman (2021b) find that trans speakers' /s/ pronunciations tend to be closer to the center of the spectrum, not as strongly gender-differentiated as cis speakers'.

Jessee and Calder (2025) analyze this through the lens of the cisgender listening subject, seeing that rather than basing perceptions on transgender speakers' actual speech, "the cisgender listening subject, which expects voices to conform to normative standards of gender appropriateness, places even stronger standards of gender normativity on trans speakers than it does on cis speakers" (9-10). Here, the cisgender listening subject will only accept an idealized highly-conforming trans person, and construes real trans speakers as inadequately performing their genders. Because we do not live up to this idealized standard, cis perceivers will misgender trans speakers, inevitably hearing them as the gender they were assigned at birth, rather than their actual gender. In combination with the $n + 1$st order of meanings for /s/, this means that attempts to move towards fulfillment of hegemonic gendered /s/ pronunciations may also be counterproductive, being misrecognized as indexing gay and lesbianness, rather than transfemininity and transmasculinity respectively. Here, trans speakers are 'damned if they do, damned if they don't' – if a trans woman does not front /s/, her transness may be questioned, and she may be heard as a man; if she fronts /s/, she may be construed only as a gay man, not a trans woman. In this difficult indexical situation, in which a trans speaker knows their cisgender listener may subject them to this hegemonic indexical field which excludes them, there are theoretically two options; defend oneself by moving towards adherence to these norms, or to refuse to do so, liberating oneself from these norms but risking negative judgments. As mentioned above, transfeminine speakers studied by Heffernan (2023) discuss both of these options, acknowledging the safety and possible gender-affirmation in taking up hegemonically normative femininity, while simultaneously not wishing to judge themselves by cisgender standards. Drawing upon all of this prior work on indexicalities of /s/, this thesis investigates whether transfeminine speakers shift towards hegemonically feminine fronter /s/ pronunciations with cis listeners, and what motivations might determine the choice to shift or not shift.

That individuals may style-shift /s/ pronunciations for gender-presentational and interactional indexical purposes has been attested by Benesch (2022), Mendoza (2024), Saigusa (2016), Holmes-Elliott and Levon (2017), Levon and Holmes-Elliott (2013), Parnell-Mooney (2019), Calder (2019b), and Zimman (2017c). Holmes-Elliott and Levon (2017) and Levon and Holmes-Elliott (2013), in a study of within-speaker variation in /s/ pronunciation in British reality television shows, find that in one lower-class show, cisgender heterosexual women use backer /s/ in conversation with other women, whereas in a higher-class show, cisgender heterosexual women use fronter /s/ in this context. In order to explain these results, they conclude that the interlocutor's gender may not be the best predictor, as in both shows women back /s/ in speech activities considered more threatening. As discussed below in the methods chapter, the recorded conversations I conducted attempted to

control for this stance-based shifting through the structure of two collaborative-valenced speech activities.

Moving to investigate /s/ shifting in queer speakers, Calder (2019b) finds that as SoMa drag queens get into their drag, visually transforming into their drag persona, their extremely fronted /s/ pronunciations get slightly backer to index their change in persona. Benesch (2022) finds that genderfluid people, who experience different genders at different times, shift /s/ to index these shifts in gender, but that these shifts do not necessarily correlate with typical scales of gender such as masculinity vs. femininity, but rather with personal, alternative scales such as 'flannel' vs. 'glitter'. Mendoza (2024) presents a case study of a drag queen in Wynwood, Miami, who uses fronter /s/ pronunciations when discussing the aspects of their drag taken from white, higher-class femininity, but uses backer /s/ when discussing their more anti-normative *chonga* drag persona. *Chonga* is a queer subversive Latinx femininity which may look like "unshaved body hair and beards, as well as hoop earrings and dramatic lip liner" (307). The polyvalence of this queen's drag persona is indexed through these changes in /s/. In a similar way to the findings of Calder (2019b) and A. J. Steele (2022), this queen reinterprets the indexicality of fronted /s/, using it not to index femininity in general, but rather to index a specific drag satire of white cisgender heterosexual higher-class femininity. In this way, meanings evolve outside of the hegemonic field.

I choose to focus on /s/ for my study, rather than other linguistic forms bearing social indexicalities, because of the theoretical depth of the research that has already been conducted on /s/; as reviewed above, this research has captured a nuanced and detailed view of the hegemonic indexical field of /s/, how it is marginalizing, and how those who are marginalized by it reinterpret and construct its meanings subversively within their communities. However, in the /s/ literature, as well as in sociolinguistic research more broadly, a gap I address is the dearth of research looking specifically at situations of indexical constraint, analyzing how marginalized speakers navigate listeners' indexical fields which marginalize them. Comparing transfeminine individuals' /s/ pronunciations in conversations with another transfem, where they can adopt their own indexicalities more freely, against their /s/ pronunciations in conversations with a cis woman, where they are forced to navigate the hegemonic indexical field, is illuminating here. More broadly, this research also addresses the absence of within-speaker style-shifting research on transfeminine speakers– although there have been studies of within-speaker variation in transmasculine people (eg. Zimman 2017c; Parnell-Mooney 2019), nonbinary people (Gratton 2016), and drag queens (Calder 2019b; Mendoza 2024), there have been no such studies focused on how transfeminine speakers speak differently in differing situations. Of specific interest is the question of individuals' navigation of the $n+1$st indexical order of /s/ frontness linked to gay-masculinity; whether transfeminine speakers understand this to be a possible judgment of the cisgender listening subject, and whether this is taken into account in the choice to shift /s/. In sum, this thesis pursues the research question:

> Do transfeminine speakers shift towards fronter /s/ pronunciations with a cis woman listener versus with a trans woman listener? What factors determine this choice?

Based on the literature reviewed, I predict:

1. Speakers who express greater feelings of danger, discomfort, dysphoria, or desire to pass around cisgender people are more likely to shift;
2. Speakers who distance themselves from hegemonic femininity are less likely to shift;
3. and speakers who express concern with gay-male misrecognition are less likely to shift.

These predictions are based on the literature reviewed above, primarily the structure of the hegemonic indexical field of /s/ it has established. Because of transfeminine people's awareness of the cisgender listening subject's application of hegemonic norms, I expect that transfeminine speakers' (conscious or unconscious) decision making around /s/ shifting will be based upon their relationship with the meanings of /s/ in the hegemonic field, as well as their relationship with hegemonic feminine norms itself; if speakers distance themselves from hegemonic femininity, or disregard the judgments of the cisgender listening subject, they are more likely to refuse to take up a fronter /s/ with a cisgender listener. However, if speakers feel more threatened by these norms, feeling greater pressure to conform to hegemonic femininity, they are more likely to shift towards fronter /s/ pronunciations in self-defense, and to match up to the hegemonic standards. Yet, if speakers are concerned with being mistaken for a gay man via their voice, they may feel that, even if they would be able to benefit from indexing hegemonic femininity, /s/ is not an available resource through which to do so.

Chapter 2

Methodology

In order to investigate whether and how transfeminine individuals change their pronunciations of /s/ based on the trans status of their conversation partner, and to be able to interpret the possible motivations behind these patterns of change, I recruited four transfeminine individuals to be speakers and two women, one cisgender and one transgender, to be conversation partners (CPs). Each speaker participated in a one-on-one audio-recorded casual conversation session with each of the conversation partners, then completed a demographic survey, then participated in an audio-recorded semi-structured interview with me about their transgender experience and their relationship with their voice. This order of study activities was selected to minimize priming effects, since it is possible questions asked in the survey and interview could affect subsequent linguistic behavior.

2.1 Participants

All participants were students of Reed College between the ages of 18 and 25, recruited by flyers, word-of-mouth, and emails to the Reed College student body mailing list and to individual friends and friends-of-friends. All speakers identified as transfeminine, and both conversation partners identified as women; one identified as cisgender and one identified as transgender. Speakers are referred to by animal-name pseudonyms, and the conversation partners are referred to as 'the Cis CP' and 'the Trans CP' respectively.

2.1.1 Survey

On the survey, participants were asked to provide any and all gender, sexuality, and race labels they identify with, as well as their pronouns. With respect to gender identity, I additionally asked participants to rank on the survey how much they generally identify with femininity and masculinity on scales of 0 to 100. However, I decided not to make use of these rankings for my analysis, as most of the participants seemed to find them uncomfortable and difficult to answer meaningfully. For the participants who were not the Cis CP, the survey also asked whether the

participant was currently on hormone replacement therapy (HRT) and for how long they had been on it, whether they had received voice feminization therapy and whether they had received voice feminization surgery. All of the trans participants were on HRT, and no participants had received voice feminization therapy or voice feminization surgery. Questions were also included to understand speakers' linguistic backgrounds– what languages they speak, whether they have primarily spoken English throughout their life or have had periods of primarily speaking other languages, the region(s) in which they grew up, and whether they had received non-gender-related speech therapy. Survey questions are provided in appendix A.

2.1.2 Interviews

After conversation sessions and surveys were completed, I also conducted 20-30 minute audio-recorded semi-structured interviews with the speakers. As a starting point, I asked questions intended to elicit background on each participant's individual relationship and history with gender and sexuality beyond the labels collected in the survey. From here, I hoped to see how participants position themselves with respect to cisgender norms and hegemonic femininity, and how this relationship may have changed and developed over the course of transition. In discussing sexuality, I hoped to gauge speakers' history of identifications and relationships with sexuality labels, especially whether they had previously identified as a gay man. The discussion that followed from these questions tended also to address the question of 'outness'; what this meant for each speaker, and how private their trans and queer status was. After this, I moved to asking questions about participants' relationships with their voices. Some of these questions were modeled after those used by Heffernan (2023) to investigate speakers' understandings of the cisgender listening subject, asking speakers to talk about how they feel their voice is perceived by cis listeners and whether they might speak differently around different people. I did not directly ask speakers whether they were concerned that their voice would be mistaken as a gay man's voice, instead allowing the topic to naturally arise if it was salient to the participant. I also included in this portion questions about sexuality and the voice, such as "Do you feel that your voice sounds lesbian?", to further clarify the role sexuality may play in determining /s/ pronunciation and shifting behavior. A full list of questions is provided in appendix B. Participants' demographic information and gender background from survey responses and interviews are summarized in tables 2.1.2 and 2.1.2.

2.2 Conversations

Before the survey and interviews, speakers participated in two 20-30 minute one-on-one audio-recorded conversation sessions, one with each of the conversation partners. As these recordings would be subject to acoustic analysis of /s/, they were conducted in a sound-isolated recording booth using clip-on lavalier micro-

Participant	Gender	Sexuality	Race	First Language
Mantis (she/her)	(Trans) Woman, (Trans) Girl	Queer (Prefers to date Trans men and Nonbinary People)	Asian American	English
Alpaca (she/it)	Woman, Girl	Lesbian	White	English
Squirrel (she/her)	Transfem, Trans Girl	Bisexual (Prefers to date Fem People)	White	English
Jellyfish (she/her)	Woman	Lesbian	White	English
Trans CP (she/her)	Woman	Ace spectrum, Demisexual	Han Chinese	Mandarin
Cis CP (she/her)	Woman	Lesbian	White	English

Table 2.1: Participants' Demographic Identity Information, from survey and interviews

phones (on the speaker as well as the conversation partner) connected to a Zoom h4n recorder. After the recording was started, I left the booth, and returned once time was up. Participants were asked to pay attention to the time on their phones or computers, and spend the first 5-10 minutes of the recording collaboratively playing a word game of their choice (eg. a crossword). This prompting was intended to get participants comfortable talking with each other on the recording and to take up a collaborative and friendly stance towards one another. Controlling for stance in this way was intended to isolate stance-related variation in /s/, as indexicalities between /s/ and confrontational/threatening stances have been found (see Holmes-Elliott and Levon 2017). After playing the game, speakers were asked to spend the rest of the time of the recording, another 15-20 minutes, getting to know each other, trying to find things in common, and talking about shared experiences. Since Alpaca and the Trans CP already knew each other, they were asked to catch up and chat as they would normally. The rest of the speakers and conversation partners did not know each other. The second portion of the conversation was intended to elicit free-flowing casual talk while maintaining similarly friendly and comfortable stance orientation. Although there were some moments of awkwardness, conversations did not include disagreements, arguing, or other confrontational elements and primarily maintained a lighthearted and friendly tone. Topics were primarily light and casual, with participants talking about common interests such as movies or chemistry, or common experiences as students at Reed. Additionally important to note is that topics were generally sim-

Participant	Time Out	Time on HRT
Mantis	∼3yr	1y8mo
Alpaca	∼4yr	3yr9mo
Squirrel	9mo (primarily online)	3mo
Jellyfish	>3yr	∼2yr
Trans CP	4yr	3yr

Table 2.2: Participants' Transgender Background Information, from survey and interviews

ilar between conversations with the Cis CP and with the Trans CP; speakers did not talk with the Trans CP about gender or being trans (I did not instruct them for or against doing so; this topic just turned out not to come up).

2.3 Analytical Approach

In order to answer the first portion of my research question – whether transfeminine speakers shift towards fronter /s/ pronunciations with a cis woman listener versus with a trans woman listener – I take a quantitative empirical approach. I operationalize spectral center of gravity (CoG) of /s/, measured in Hz, as the dependent variable, and conversation partner (Cis CP vs. Trans CP) as the independent variable. In order to measure CoG of /s/, I produced a time-aligned plain-English transcript of each recording. As a first pass, I used Deepgram's Nova 3 speech recognition model, specifying that it should transcribe utterances (rather than paragraphs or one unsegmented block of text) and annotate each utterance with who it was spoken by: the speaker or the conversation partner. I then used a Python script to convert the output of this model into a time-aligned transcription file format (TextGrid) usable by the linguistic acoustic analysis software Praat (Boersma and Weenik 2023). I then manually corrected the transcripts in Praat, fixing the mistakes the automated speech recognition model made and ensuring accuracy. I then used the Montreal Forced Aligner with the included English acoustic model and dictionary to automate the production of a time-aligned phonetic transcript, which would specify, for both the speaker and the conversation partner, the start and end time in the recording of each /s/ pronunciation (McAuliffe et al. 2017). From there, I used DiCanio's (2021) Praat script to measure the CoG of each of these /s/ pronunciations in each of the recordings for the speakers and conversation partners. It was important to measure not only the /s/ pronunciations produced by the speaker participants, but also those produced by the conversation partner participants, in order to determine the extent to which the speakers may have been accommodating to the /s/ pronunciations of the conversation partners (and vice versa). I used the statistical analysis software R (R Core Team 2025) to combine, statistically analyze, and visualize the data.

In determining the second half of my research question – what factors determine

whether speakers shift to a fronter /s/ pronunciation with the Cis CP – I take a qualitative approach. I use interview responses to qualitatively assess the extent to which each speaker aligns with the three criteria in my predictions, quoting and discussing instances in which speakers:

1. express feelings of danger, discomfort, dysphoria, or desire to pass around cisgender people;
2. differentiate themselves from, or otherwise claim to disregard hegemonic femininity;
3. and express concern with gay-male misrecognition.

This approach allows me to analyze speakers with respect to their indexical goals, rather than simply comparing them against a white cisgender heteronormative standard which will inevitably find them somehow faulty (see Calder and A. Steele 2025; Zimman 2021a). Instead, I aim to understand the connections between speakers' understandings of their voices and their vocal behavior, underlining instances of strategic deployment of semiotic resources, as well as the oppressive limitations imposed by cisnormativity.

Chapter 3

Results

	Number of /s/ tokens		Duration of Recording	
Speaker	w/ Trans CP	w/ Cis CP	w/ Trans CP	w/ Cis CP
Mantis	177	143	21m48s	14m31s
Alpaca	476	244	30m56s	21m53s
Squirrel	388	401	25m04s	29m49s
Jellyfish	444	283	26m27s	20m08s

Table 3.1: Table of number of /s/ tokens measured and duration of each recording.

	Mean /s/ CoG (Hz)			
Participant	Overall	w/ Trans CP	w/ Cis CP	p-value
Mantis	7231	6638	7964	<0.001
Alpaca	5758	5899	5483	<0.001
Squirrel	6357	6345	6369	0.8263
Jellyfish	6154	6039	6334	<0.01
Trans CP	5133	N/A	N/A	N/A
Cis CP	6835	N/A	N/A	N/A

Table 3.2: Table of each speaker's mean /s/ CoG each speaker, overall as well as separated by conversation partner (CP) with p-values (I take p-values of <0.01 to indicate a statistically significant shift.)

Table 3.1 lists the duration of each recording and the number of /s/ tokens produced by the speaker participant in each recording. Table 3.2 and figure 3.1 present an overview of participants' /s/ pronunciations. Zimman (2017c) reports that so-called 'reference' studies on white English speakers presumed to be cisgender and heterosexual report a 'typical' /s/ CoG range of 6,400Hz–8,500Hz for women and 4,000–7,000Hz for men. That only Mantis and the Cis CP fell inside the range expected of women, and that only Mantis fell outside the range expected

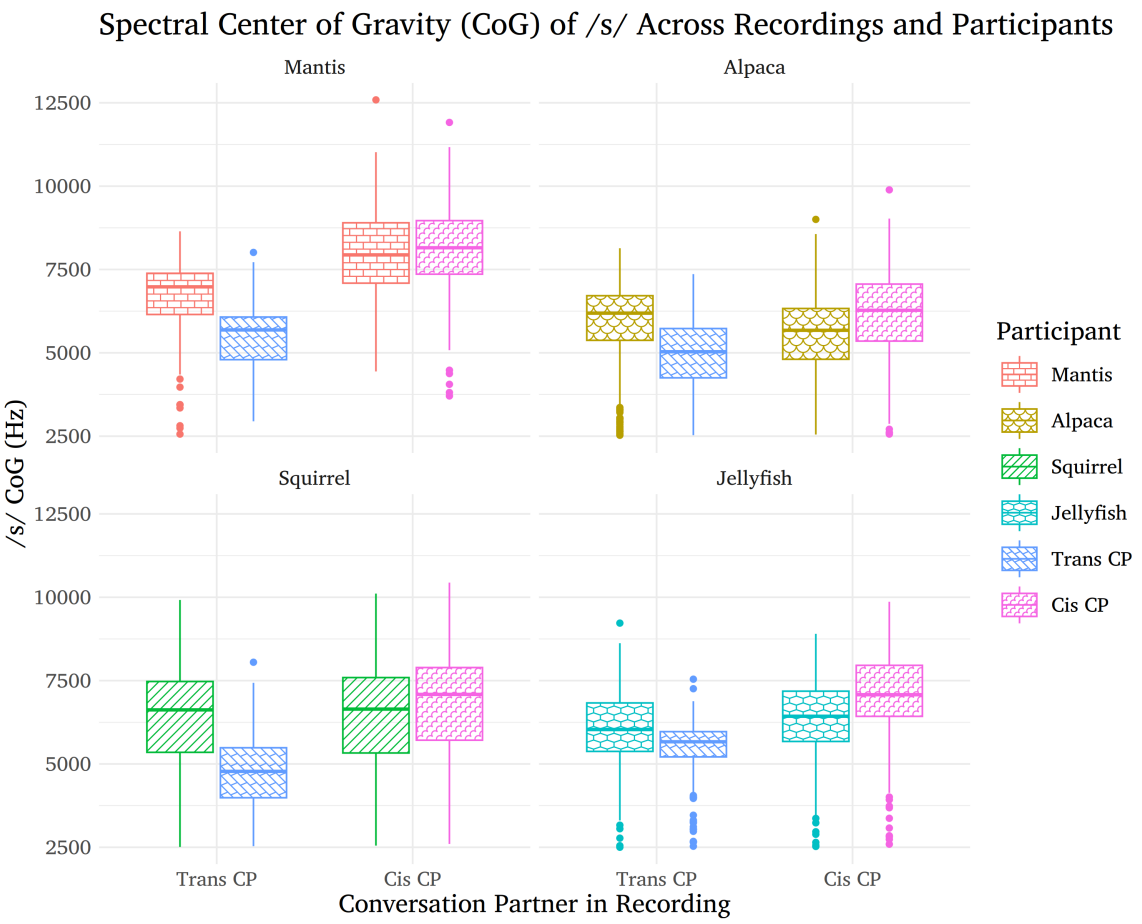


Figure 3.1: CoG of /s/ pronunciations in each recording, by each speaker.

of men, despite the fact that all of my participants identified as women or girls, points towards the inadequacy of these 'reference' values outside the cisgender heterosexual white population (recall that the Cis CP is a lesbian). I argue that this discrepancy should not be read as an indication of inaccurate or flawed measurement methodology, however, as my speakers' means do align with previous measurements for queer and trans women: the white trans women and cis lesbians Podesva and Van Hofwegen (2015) study in Redding, California exhibit mean /s/ CoG 6,000Hz and 6,350Hz respectively, and the white trans women Hazenberg (2012) studies in Ottawa, Ontario exhibit a mean /s/ CoG of 6,600Hz. These measures align much more closely with my speakers, with the exception of the Trans CP and Mantis, who occupy the extremes of my data. I interpret this general correspondence of my data with previous studies of trans and queer women as confirming the validity of my measurements.

With respect to shifting, three of my four speakers, Mantis, Alpaca, and Jellyfish, exhibit statistically significant shifts in /s/ CoG from the recording with the Trans CP to the recording with the Cis CP, and one speaker, Squirrel, did not. Of the three speakers who shifted, two, Mantis and Jellyfish, shifted in the direction

of fronter /s/ pronunciations producing higher CoGs when talking to the Cis CP. This accommodation shifts their speech into closer alignment with hegemonic expectations of the feminine voice. The other speaker who shifted, Alpaca, shifted in the opposite direction, shifting to backer /s/ pronunciations with the Cis CP, moving away from the hegemonic norm in this context. With respect to the degree of shifting, Mantis exhibited the largest shift with a difference in means of >1000Hz, while Alpaca and Jellyfish shifted more subtly. Squirrel, who did not shift, maintained the same (relatively fronted) /s/ pronunciation across both conversations with a mean CoG of 6357 Hz.

In the sections that follow, I consider each speaker individually, qualitatively assessing their commentaries in the interview for the attitudinal factors listed above (section 2.3) in reference to their /s/ shifting behavior. In addition to assessing these attitudinal factors, I note other possibly-relevant or explanatory aspects of each speaker's experience which were raised in my interview with them.

3.1 Speaker 1: Mantis

Figure 3.2 zooms in on Mantis' /s/ pronunciations. Her /s/ behavior is distinct from the other speakers' – her shift is the largest (with a difference in means greater than 1000Hz), and her overall mean CoG is the highest. As mentioned above, her /s/ pronunciations fall within the hegemonically-expected 'reference' range for cisgender women with an overall mean of 7231hz. In my interview with her, Mantis told me that she has "had so many people come and tell [her] that they thought [she] was a cis woman with [her] voice"; it is possible that the frontness of her /s/ pronunciations contribute to these perceptions. She notes that these instances of having a cisgender-passing voice are the reason that she "used to worry a lot about [her] voice, but now it doesn't bother [her] as much." Here, she constructs a narrative about the development of gender confidence over the course of transition. Alpaca, Jellyfish, and the Trans CP – the other participants in the study who had been out as trans women for multiple years – all related similar stories of development in my interviews with them. Although this may read as an expected part of a gender transition, it is nonetheless important to note, because for Mantis and Jellyfish, part of these narratives was the development of a greater comfort with deviation from hegemonically-normative standards of femininity. Mantis notes:

> "I don't necessarily consider myself to be a woman in a way that a binary woman is a woman, but I also don't say I'm nonbinary, because I very much am like 100% woman, it's just, not in anybody else's way."

Here, Mantis not only distances herself from hegemonic femininity, but positions herself outside of it. Understanding her womanhood as unique to herself, Mantis asserts that she does not have to adhere to 'anybody else's way' in order to be '100% woman'. Here, Mantis emerges as a counter-example to one of my predictions: she shifts, despite distancing herself from cisgender norms.

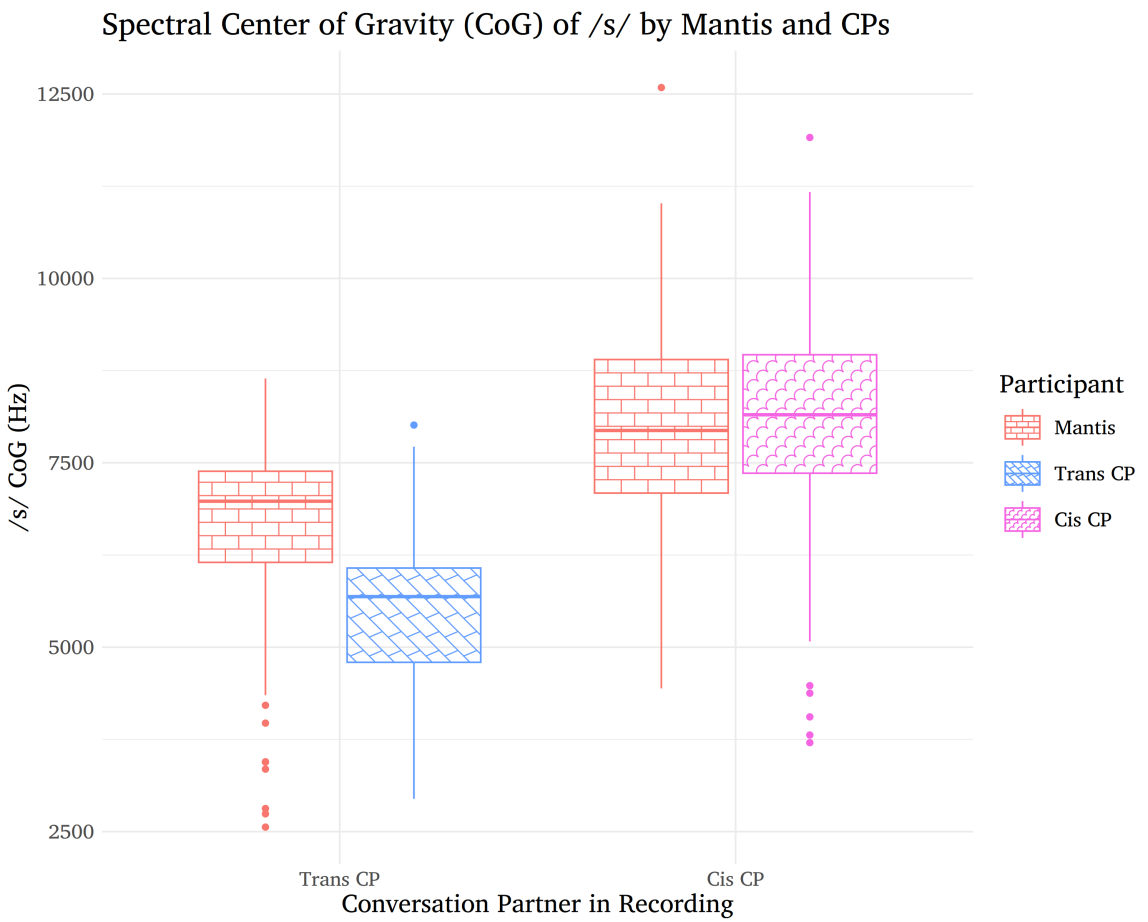


Figure 3.2: CoG of /s/ pronunciations in each recording containing Mantis, her pronunciations and the pronunciations of the conversation partners.

An additional counter-example provided by Mantis relates to the issue of gay-masculine misrecognition. She attests that sounding like a gay man is a concern, but this does not lead her to avoid fronter /s/ pronunciations with the cisgender listener as I predicted. When asked how her experience with her own voice has developed over time, she began her answer by noting that she has "always had like 'gay voice' or like feminine voice". When I asked whether her voice dysphoria was primarily the concern of not sounding feminine enough, she replied:

> "... it was, more specifically, like 'oh I just sound like a gay man'. And then there would be gay men who have gay voice ... and I'd be like 'oh I can't distinguish my voice from theirs', and it's just very frustrating, and I don't want people to think like that of me."

From this answer, it was clear that concerns about gay-masculine misrecognition represent an important part of Mantis's relationship with her voice. I discuss the possible interpretations of this apparent mismatch in attitude and behavior in the following chapter. At this point is is sufficient to note that Mantis stands here as a

counterexample to my prediction that speakers concerned with misrecognition as a gay man would avoid fronter /s/ pronunciations with the Cis CP.

On the other hand, Mantis did provide evidence to support my first prediction, that speakers who reported greater feelings of pressure to conform or 'pass' with their voice in the presence of cisgender listeners would be more likely to shift to a fronter /s/ pronunciation with the Cis CP. In our discussion of voice dysphoria, Mantis mentions:

> "in conference sometimes ... I'm like, I don't know, maybe I shouldn't speak, because the person who just spoke before me just sounded like, so, woman, and now if I speak, people are gonna think I sound like a man. ... especially with cishet[1] women ... it's almost like a competition."

In these moments, Mantis feels that cisgender women's speech becomes the standard for feminine speech, and that the differences between her voice and the voices of cis women around her mark her speech as deficient and male. In order for feminine speech to be a 'competition', not only does there have to be a unitary way to 'win' – the cisgender norm – but further, she and her listener both have to make effort to change their speech towards this norm. This is backed up by her behavior, as well as the behavior of the Cis CP; both of them shift towards fronter /s/ pronunciations with each other. In her conversation with Mantis, the Cis CP's mean /s/ CoG is 8009Hz, significantly different from her mean /s/ CoG with the other speakers (6642Hz; $p < 0.0001$). Under an analysis which centers accommodation, it is unclear who is accommodating to whom here; however, in light of Mantis' notion of a competition, and the known indexicalities of /s/, this behavior can be understood as both speakers accommodating towards the hegemonically-expected feminine /s/ pronunciation.

3.2 Speaker 4: Jellyfish

Jellyfish, like Mantis, shifted towards fronter /s/ pronunciations when talking to the Cis CP, as shown in figure 3.3. Compared to Mantis, however, Jellyfish's frontwards shift is smaller, with a difference in mean CoG of 300hz, and her overall mean CoG of 6154hz is lower, closer to Squirrel and Alpaca. In my interview with her, Jellyfish emphasized the importance of lesbianism to her identity and her relationship with her voice, saying that "being a lesbian impacts the way that [she] move[s] through the world as a woman more than being trans does in a lot of situations," and that "[she] can sound like a boy as a lesbian, with the way that [she] present[s], and have people just be like 'oh that's like a weird dyke who smokes cigarettes'," which she considers to be an accurate description of herself. In this way, it makes sense to read her mean /s/ CoG as similar to that of the lesbian speakers Podesva and Van Hofwegen (2015) studied in Redding, California whose mean /s/ CoG was 6300Hz.

[1] 'Cishet' is a shortened form of 'cisgender heterosexual'.

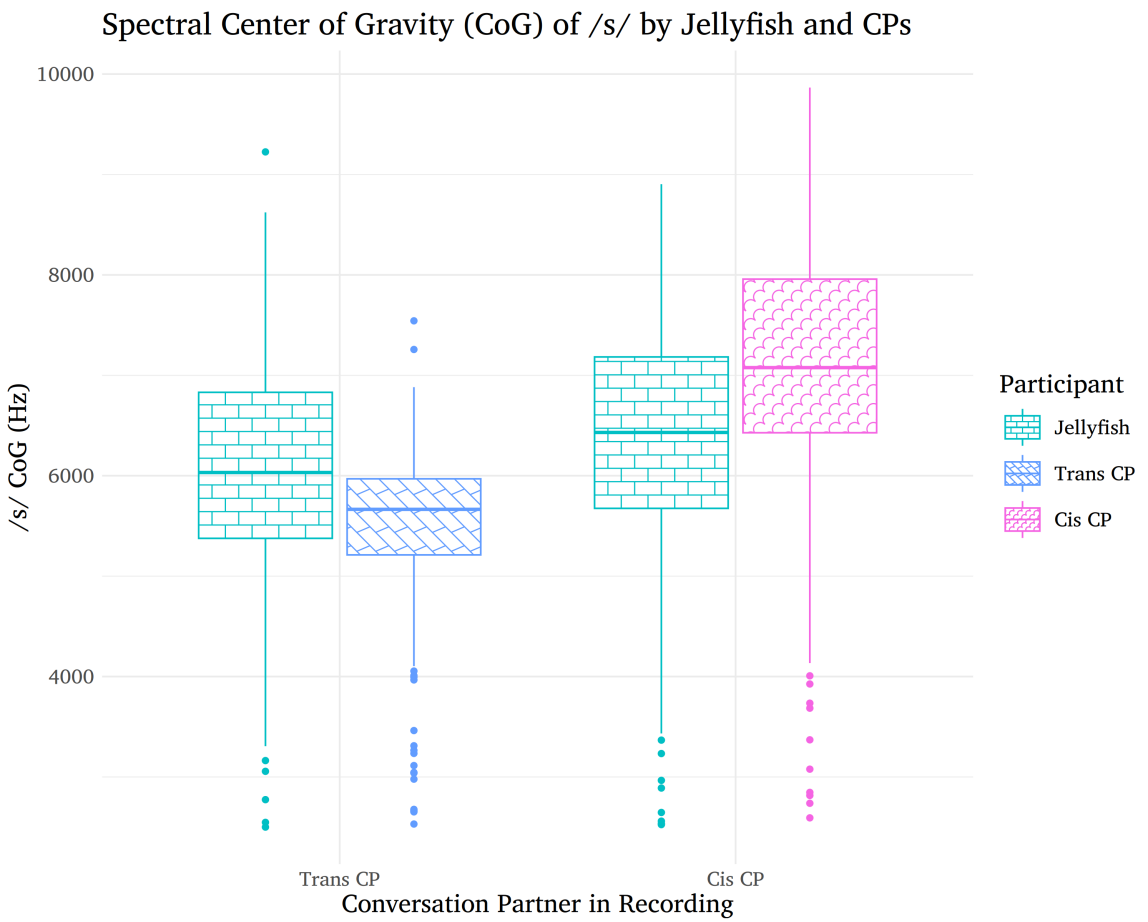


Figure 3.3: CoG of /s/ pronunciations in each recording containing Jellyfish, her pronunciations and the pronunciations of the conversation partners.

Similar to Mantis, Jellyfish lays out her transition as a process of increasing confidence in her gender and increasing ability to disregard hegemonic feminine norms. For Jellyfish, this takes the shape of the development of a lesbian boy identity against earlier internalizations of ideological transphobia. She notes:

> "When I was in high school ... I would like look at all this like TERF[2] propaganda and more or less like fashion myself to be the exception to like the narrative around trans people."

She later notes that this caused her to become "kind of a psychopath, about like, presentation", paying obsessive attention at that time to "do like everything possible to fool people into thinking [she] was a cis woman". This was also the period where she notes:

[2]'TERF' is an acronym for 'trans-exclusionary radical feminist', a label for a transphobic ideology which attempts to justify itself through appropriation of feminism (see the discussion of gender-critical feminism and TERFism in Barrett and Hall 2024).

> "I realized that women whistle on their 's's way more, and I remember like consciously changing how I speak to like whistle on the 's' more."

This level of conscious awareness of /s/ fronting is striking, and very interesting on its own. But even as an example of heightened awareness of hegemonic feminine norms and concern with having a 'passing' voice, this can be read as providing evidence for my prediction that speakers who voice greater concern with cisgender perceptions will shift to fronter /s/ with the Cis CP. However, it is important to note that Jellyfish situates this quite clearly as an aspect of her past more than her present.

By contrast, Jellyfish's present is characterized by a rejection of strict conformance to a singularly and hegemonically feminine gender and expression. This rejection allows her to feel comfortable engaging in practices which may be hegemonically labeled masculine without judging herself as insufficiently performing womanhood. She expresses this confidently:

> "I don't think it benefits me in any sort of way to like restrict myself like that... people are way to serious about all this shit, ... like, if I'm dressed more masculine in a certain day I'll like listen to different music and I'll like talk differently and I'll move differently and I just think that's fun!"

When considered in light of my prediction that speakers who distance themselves from hegemonic feminine norms will not shift to fronter /s/ with the Cis CP, Jellyfish may be read as an additional counter-example; she shifts, despite this distancing.

Since Jellyfish and the Cis CP were both lesbians, is also possible here that Jellyfish's shift could be a move to create similarity between herself and her listener, a move of accommodation. Here, Jellyfish could be seen as enacting a sort of harmonizing gender fluidity, in which she indexes the gender and sexuality of people around her in order to create social closeness. This view is also supported by her comment that she "love/s/ talking like a boy when [she's] interacting with other boys".

Like Mantis, Jellyfish serves as a counter-example against my prediction that speakers who expressed concern with gay-masculine misrecognition would not shift to fronter /s/. She discussed this issue with me in some detail, clarifying points which are helpful in the interpretation of both her and Mantis' behavior. She presents the issue of 'sounding like a gay guy' as of particular concern in interaction with cis women, and as an unintentional habit from the past, noting that:

> "It is like, annoying at times when I like, default back to talking like a gay guy around cis women ... more than sounding like a boy sounding like a gay guy is like, terrifying, because there's this relationship that, it's mostly cis women, have with like trans women where you will be accepted and tolerated as like a gay guy but anything beyond that, if you like, don't know your place more or less, you're getting into dangerous territory."

Jellyfish clarifies here that 'sounding like a gay guy' is not only a concern because of the inaccuracy and denial of her gender, but further that it represents a dangerous type of transphobia which hides itself behind an apparent friendliness towards cisgender gays and lesbians. In the phrasing 'default back to', she situates 'talking like a gay guy' as a habit from her past. She mentioned in our interview that she identified as a gay boy for a period of time before realizing she was trans; it is possible that speech habits from that time remain in her speech. I discuss the issue of listeners' personal history in relation to the question of gay male recognition, as well as the issue of intentionality and difficulty of changing how one speaks, in detail in the next chapter.

3.3 Speaker 2: Alpaca

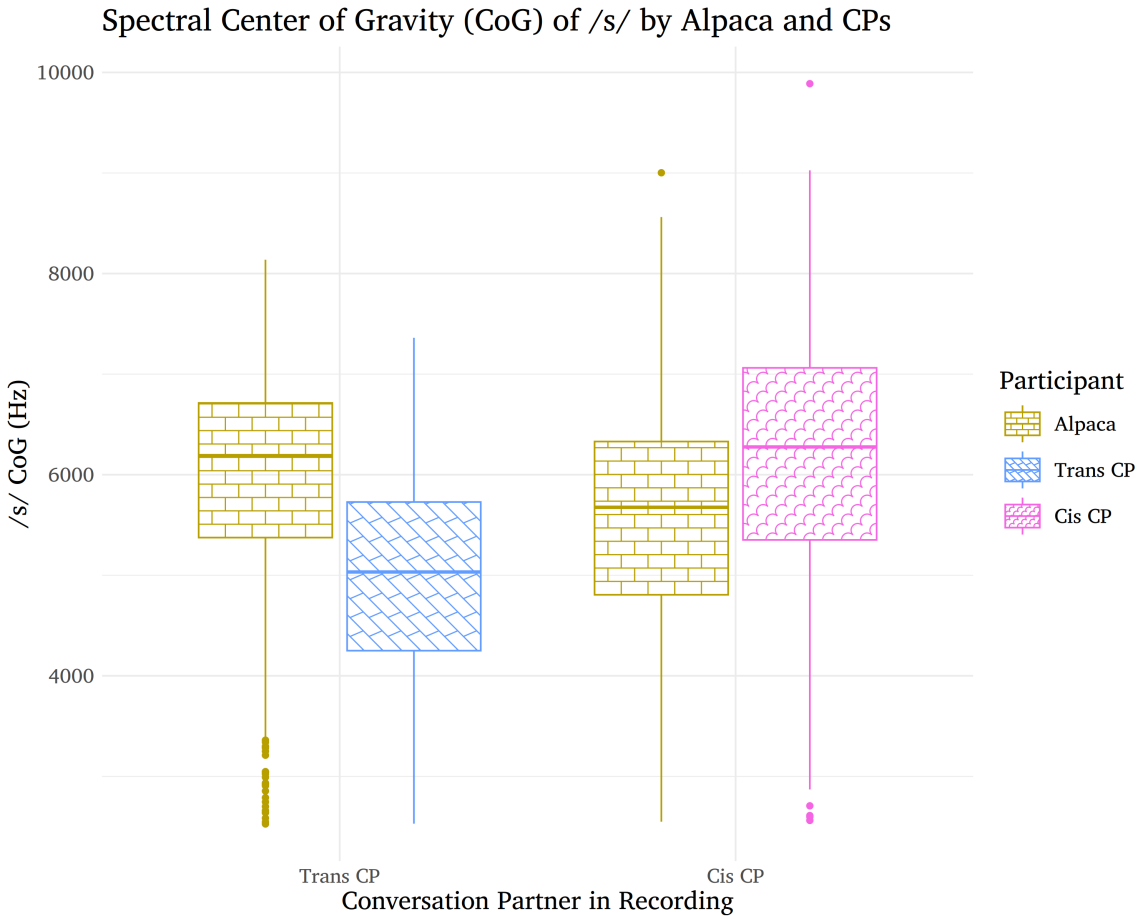


Figure 3.4: CoG of /s/ pronunciations in each recording containing Alpaca, her pronunciations and the pronunciations of the conversation partners.

Alpaca's /s/ pronunciations are shown in 3.4. Alpaca is my only speaker who shifted her /s/ pronunciations to be backer with the Cis CP, rather than fronter;

her mean CoG reduces by 400Hz. Overall, her /s/ pronunciations are somewhat backed, with a mean CoG of 5758Hz. This is the second lowest mean among my participants, only higher than the Trans CP's (5133Hz).

Unlike Mantis and Jellyfish, Alpaca did not mention concern with sounding like a gay man in my interview with her. In our discussion of sexuality, she also did not mention previously identifying as a gay man. In my initial predictions, shifting behavior like Alpaca's would have been most expected from speakers who voiced concerns with gay-male misrecognition, shifting away from fronted /s/ with the cisgender listener to avoid this misrecognition. That Alpaca – who shifts in this manner – does not have this concern (or at least does not voice it in my interview with her), while Mantis and Jellyfish – who shift in the opposite way – do, encourages a reinterpretation of my prediction with regards to concerns with gay-masculine misrecognition. This reinterpretation is elaborated in the following chapter.

With respect to relationship with hegemonic femininity, the other factor my predictions associate with a lack of fronting around the Cis CP, Alpaca seems to be more neutral, not strongly differentiating or aligning herself from hegemonic conceptions of femininity. Similar to Mantis and Jellyfish, Alpaca also constructs a narrative of developing gender confidence as her transition progresses. However, the valence of this progression is somewhat different for her than the former two speakers. For Mantis and Jellyfish, the development of gender confidence involved a movement away from an initial feeling of pressure to strictly adhere to hegemonic femininity, and a gradual reclamation of behaviors which may hold hegemonic masculine associations as valid parts of a woman's identity. Alpaca describes the progression of her understanding of her gender differently:

> "I think it's been a process of just sort of like getting more comfortable with femininity, because not wanting to be masculine, is, in my thought process, is kind of different from, ... coming to terms with the fact that I do like femininity and that I am comfortable with it and that it is something I can enjoy and be comfortable in and have as my own part of my identity ..."

Here, it seems that Alpaca's transition journey involves more of a gradual move towards comfort with elements of trans-affirming femininity, rather than a gradual move away from strict adherence to hegemonic femininity. Similar to the case with gay-male misrecognition, it is possible that part of this development for Alpaca does include distancing herself from strict adherence to hegemonic femininity and that she simply did not mention so as explicitly as Mantis and Jellyfish. However, from my interview with her, I am unable to claim with certainty whether this is the case, and whether this could explain her backing of /s/.

With respect to the degree of concern with passing and cisgender perceptions, Alpaca expressed similar feelings as my other speakers. Although she does not word this quite as strongly as the prior two speakers, not expressing frustration or feelings of danger in connection to cis perceptions, she notes

> "when I'm around other trans people I kind of feel less pressure ... to try to present perfectly femininely, like I feel less pressure to like pass ... in class or around cis people I might try to have a more feminine voice."

Counter to my prediction, this 'pressure to pass' does not result in a shift towards fronter /s/ pronunciations with the Cis CP for Alpaca.

From the perspective of my three attitudinal predictions, I am unable to provide a solid explanation for Alpaca's shift to backer /s/ pronunciations with the Cis CP. However, there are two possible confounding factors which may contribute to an explanation. First, Alpaca notes that she was not completely sure whether the Cis CP was cisgender. This was a result of my choice not to explicitly inform speakers of the trans status of the conversation partners, in order to reduce priming effects. Although she did suspect that the Cis CP may have been cis, she also believed it was possible that the Cis CP was a highly passing trans woman. As a possible confounding factor, this may also play a role in explaining Alpaca's difference in behavior from the other speakers. Second, unlike the other speaker participants, Alpaca was friends with the Cis CP, another possible confounding factor.

3.4 Speaker 3: Squirrel

Squirrel stands out as the only speaker who did not shift her /s/ pronunciations, maintaining a mean CoG of 6357Hz with both of the conversation partners (as shown in figure 3.5). She is also the only speaker in the early stages of her transition. She realized she was trans less than a year ago, and has not publicly come out yet; only online friends and those close to her know that she is trans. In my interview with her, she explained that she wished to physically transition through HRT before socially transitioning, and that she "didn't feel comfortable enough yet to" come out publicly. She plans to come out after about a year of HRT, and intends to work on voice training before doing so. She did participate in this study in her transfeminine identity, however, and the CPs were aware that all of the people they would be talking with were transfeminine.

At this early stage of transition, it is perhaps surprising that her /s/ pronunciations are not backer, as she is the speaker participant with the second highest overall mean at 6357Hz. Although she has not yet begun efforts on voice training, it is possible she has already taken some less conscious steps towards a feminine voice, or that her /s/ pronunciation was already in this range pre-transition. In the interview, I asked her whether she felt that her voice was already feminine now, to which she replied:

> "that's a complicated question... sometimes, sometimes not... when I was really young ... I used to be really insecure about my voice, but for the opposite reason, that I thought it was like too fem."

She later cited self-perceptions of her voice as high-pitched as the primary cause

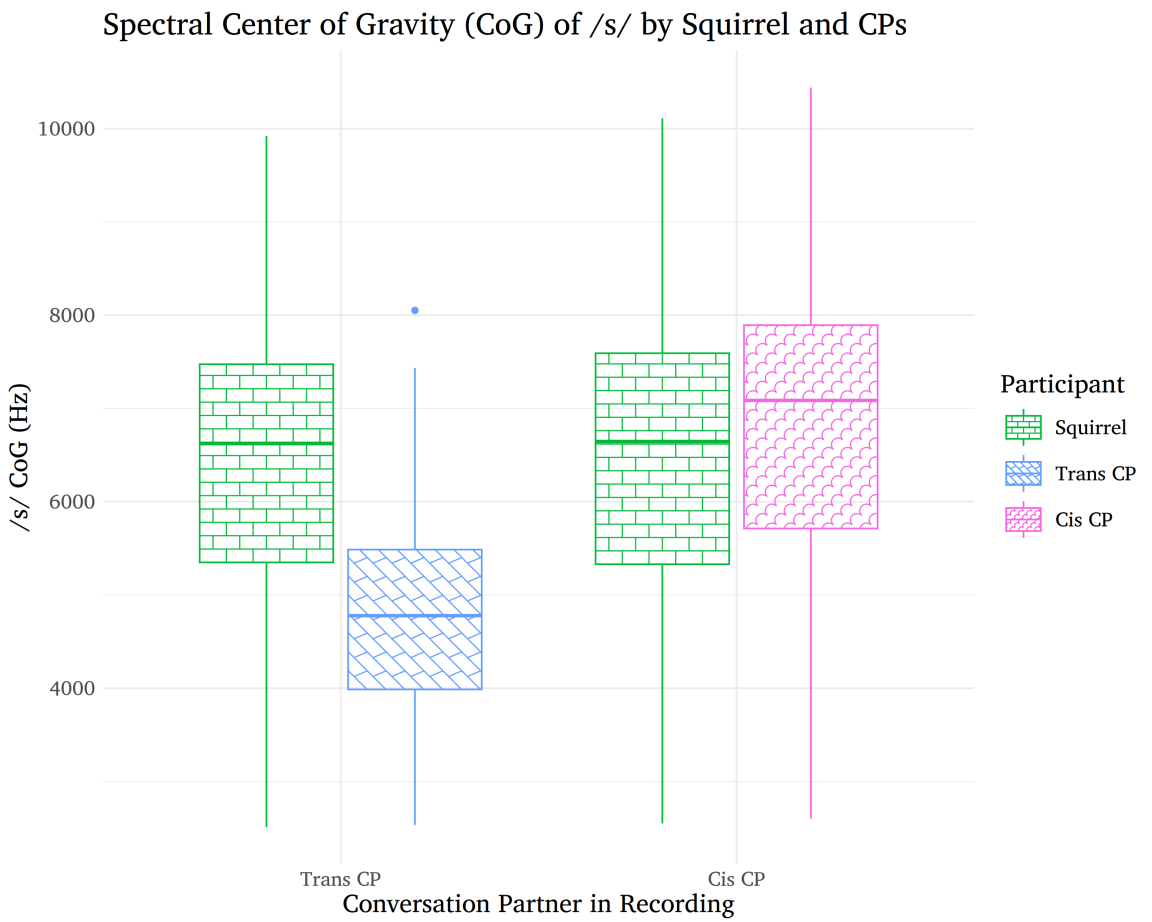


Figure 3.5: CoG of /s/ pronunciations in each recording containing Squirrel, her pronunciations and the pronunciations of the conversation partners.

for these pre-transition insecurities, but it is possible that /s/ pronunciations also played a role.

According to my predictions, speakers who distance themselves from hegemonic feminine norms and speakers who have concerns with gay-masculine misrecognition should be less likely to shift /s/ pronunciations. From my interview with her, it seems that the former could be a factor in her choice not to shift. She notes that although she does intend to move towards a more feminine presentation as she transitions, this does not equate to a desire for strict adherence to traditional hegemonic femininity:

> "I think a lot of times there's like this expectation of, if you're transfem that you're like hyper-fem, and that's not how I feel really personally, like I don't know if 'butch' is the right word for that but I definitely feel like the way that like I wanna present is not like super-hyper-fem."

Similar to Mantis and Jellyfish, Squirrel affirms here that she does not want to and does not have to strictly follow hegemonic feminine norms, and that she hopes to inhabit a femininity which may contain elements hegemonically associated with

masculinity. She explains this through clothing, mentioning that she would prefer to wear all-black than a brightly colored sundress.

Squirrel, like Alpaca, does not mention concerns about being mistaken for a gay man with the sound of her voice. In our discussion of the history of her relationship with her gender and sexuality, it seemed that she did not identify as a gay man at any point. As a result, my prediction with respect to concerns with gay-male misrecognition turns out to be the opposite of the pattern found; the two speakers who are concerned with gay-male misrecognition, Mantis and Jellyfish, shift to fronter /s/ pronunciations with the Cis CP, whereas the two speakers who do not have this concern, Alpaca and Squirrel, do not. It is additionally of note that Alpaca's /s/ pronunciations are fronter, similar to those reported for gay men by Podesva and Van Hofwegen (2015). That she does not have concerns with gay-masculine misrecognition provides evidence that these concerns may not be caused by a pre-existing speech pattern; the reverse seems to be the case. I return to the matter of gay-masculine misrecognition in the following chapter, but it is sufficient to note here that concern with misrecognition does not appear to be a factor in Squirrel's lack of shifting, counter to my predictions.

Squirrel's lack of shifting may also be interpreted as a counter-example to my prediction that speakers who express greater concern with cisgender perceptions and 'passing' would be more likely to shift. If anything, Squirrel seemed to hold these sorts of concerns and discomforts most strongly among my participants, perhaps in part due to her transition stage. When I asked her about her relationship with her voice, she began by explaining that in online voice chat interactions,

> "I like don't talk to a lot of people like that I would normally talk to if I was cis... as soon as I like basically came out to like my [online videogaming group]... I stopped talking entirely, on voice. ... I feel like it's easier for them to perceive me and like use correct pronouns like, if they don't hear me, because my, there's voice dysphoria, and like, my voice isn't like, the way I would want it to be if I could like pick it basically."

She explains further that

> "I feel like inherently when like you hear someone for the first time you kind of assign like gender in your head to like that person... if you hear a stereotypically more masc voice then you might like automatically start like 'he/him'ing someone in your head."

Here, she implicitly characterizes her voice in its current state as 'stereotypically more masc', in that she feels that not using her voice is the best way to avoid being perceived as a man in online interactions with cisgender people. For her, this falls under a general preference articulated as "the more ways I can like not be perceived, the better." In this context, it is perhaps understandable that pressure to pass does not lead to a shift in /s/, as Squirrel perceives a passing or feminine voice as not yet in reach at her current stage of transition.

Chapter 4

Discussion

In this chapter, I discuss the above results in light of the research question of this thesis raised above:

> Do transfeminine speakers shift towards fronter /s/ pronunciations with a cis woman listener versus with a trans woman listener? What factors determine whether speakers shift?

The answer to the first portion of this question seems relatively clear; Mantis and Jellyfish shift in this way. However, in order to be sure that this answer is not misleading, it is necessary to consider whether the speakers are specifically shifting in response to the trans status of the CP and the hegemonic indexical field of /s/, rather than in response to some other factor. Of concern here is the issue of accommodation; could these shifts be explained simply as the speakers pronouncing /s/ more similar to the CP they are talking with, rather than in reaction to whether the CP is cis or trans? The section that follows considers this question in detail. I find that although I cannot answer this question with certainty, it seems more likely, and more interpretively productive with respect to the second portion of the question (which factors determine whether speakers shift), that speakers are specifically reacting to the trans status of the CP and the hegemonic indexical field of /s/, rather than simply accommodating.

With respect to this second portion of the question – what factors determine whether speakers shift – I put forth an interpretation in which personal history of gay-masculine self-perception emerges as the primary factor. This factor differentiates the two speakers who shifted (towards fronter /s/ with the Cis CP) from the two who did not, but in the opposite way of my predictions. In light of this, I reinterpret gay-masculine misrecognition through the lens of personal history and salience of indexicalities of /s/. I then briefly discuss the role of race in my data, the failure of my other two predictions, and the implications of this data for the theories of misrecognition, the cisgender listening subject, and the hegemonic indexical field.

4.1 The Issue of Accommodation

Due to the small scale and scope of this study, my research design is unable to fully eliminate the factor of accommodation. In order to do so, it would either need to be the case that the two CPs have the same /s/ pronunciations – in which case accommodation could not result in any shift – or that more than one of each type of CP was included (e.g. two trans CPs and two cis CPs) – in which case shifts across listeners of differing trans status could be compared with shifts within listeners of the same trans status. Future research employing these designs would be illuminating. In the absence of these measures, it is undoubtedly the case that accommodation is taking place for Mantis and Jellyfish to some extent; their mean /s/ pronunciations fall in between the mean /s/ pronunciations of the CPs, and their shifts move them closer to the pronunciations of their CP, fulfilling the definition of accommodation. To point, though, is not whether these speakers are accommodating, but whether their motivations for their accommodation can be interpreted without reference to the trans status of the CP and the hegemonic indexical field for /s/. To answer this question, I compare the accommodation-only interpretation against the indexical interpretation, in terms of their ability to explain why Mantis and Jellyfish shift in accommodation with hegemonic feminine norms for /s/, while Alpaca and Squirrel do not.

How, then, might theories of accommodation differentiate Mantis and Jellyfish as more motivated to accommodate than Alpaca and Squirrel? Research on sociolinguistic accommodation generally considers differences between individuals' propensity towards accommodation within two frameworks, neither of which, I argue, provide clear answers here. First, based upon patterns found within the cisgender heterosexual white population, it is often generalized that women accommodate more than men (e.g. Giles 2016). As all four of my speakers identify as women or girls, this generalization would thus expect them to all be similarly inclined towards accommodation, and cannot differentiate the two pairs of speakers. Second, accommodation is considered a tactic of intersubjective adequation (e.g. Giles 2016; see Bucholtz and Hall 2005 on tactics of intersubjectivity), whereby speakers accommodate in order to create a sense of similarity and solidarity between each other. This conceptualization can be used to understand why individuals may accommodate to different degrees to different conversation partners, but is not particularly of use in distinguishing why some individuals accommodate – in general, to all conversation partners – more than others. When the CPs are considered together, it is not clear that Mantis and Jellyfish were more interested in creating solidarity between themselves and the CPs than Alpaca and Squirrel were. This might be expected if, say, there was some in-group to which the CPs, Mantis, and Jellyfish belonged, but Alpaca and Squirrel didn't. I am unaware of any way in which this could be the case. Mantis and Jellyfish did not seem to have any more in common with the CPs than Alpaca and Squirrel did – aside from Alpaca's friendship with the Trans CP, which, notably, did not result in accommodation.

At this point, the interpretation in which Mantis and Jellyfish are viewed as neutral accommodators, not reacting to the trans status of their CP or the hege-

monic indexical field of /s/, ceases to provide helpful interpretations. Although I cannot reject the possibility outright – it is always possible there is factor outside of my data which could explain why Mantis and Jellyfish are simply more inclined to accommodate than Alpaca and Squirrel – I nonetheless find it unlikely that the trans status of their CPs and the hegemonic indexical field of /s/ are not relevant to these speakers' behavior. Aside from the motivations with respect to explaining the difference between the two pairs, I find it unlikely that trans status of the CP was not relevant because of how my speakers talk about cisgender perceptions in our interviews, repeatedly considering their voices and how their voices are perceived through the lens of the cisgender listening subject.

If speakers' /s/ shifting behavior is interpreted as responding to these factors, the answer to the first portion of the research question posed at the beginning of this chapter – whether transfeminine speakers shift towards fronter /s/ pronunciations in response to whether their listener is cisgender – is that Mantis and Jellyfish do. How does this approach, then, address the issue of why Mantis and Jellyfish shift to fronter /s/ with the Cis CP and Alpaca and Squirrel do not? I take speakers' commentary on the issue of sounding gay-male as the primary explanatory factor to this question.

4.2 Motivations to Shift

In my original model, I expected Mantis and Jellyfish's expressions of concern with sounding like gay men to lead them to avoid fronting /s/ with a cisgender listener, out of concern that /s/ may be interpreted as indexical of a gay man identity. This was not borne out, but it is nonetheless the case that Mantis and Jellyfish differ from Alpaca and Squirrel on this issue – the former pair expressed these concerns, the latter pair did not. Here, I reconsider this issue in terms of personal history, exploring why speakers with a pre-transition history of a self-perceived gay-masculine voice might be more likely to front /s/ with cisgender listeners.

For Mantis and Jellyfish, I notice the issue of a perceived gay-masculine voice arises from these speakers' pasts. When asking speakers about their relationship with their voice and their gender, I framed the question in terms of histories, asking how these speakers relationships with their voices changed and developed throughout their life and transition. It is in this context that Mantis says that she has "always had like 'gay voice' or like, feminine voice"; the use of the word 'always' emphasizes that 'gay voice' was already a concern for her before transition. As I mention in Jellyfish's section of the previous chapter, she similarly situates 'talking like a gay guy' as something that she sometimes 'defaults back to'. Because these concerns existed before transition for these speakers, the notion of 'sounding gay' may have impacted speakers' habits of language use more thoroughly or for a longer period of time than being trans has.

At this point, it is important to clarify the nature of these concerns. What do these speakers mean when they ascribe gay-masculinity to their voices? According to the body of research on the notion of gay speech, and according to Mantis'

equivocation, 'gay voice' may simply be 'feminine voice' features in the context of a speaker or speech otherwise perceived as masculine, relying on the fractal-recursive structure of gendered and sexualitized sociolinguistic meanings. It is notable Mantis and Jellyfish apply these perceptions to themselves; they do not frame their discussions in terms of others saying that they sound like gay men, but rather as hearing themselves sounding like gay men. Before transition, these self-perceptions may then be conceived of as speakers' noticing femininity within themselves despite maintaining an assigned masculine identification, resulting in self-perceptions of gay-masculinity. That these concerns persist after transition for my speakers could then be considered a result of lingering masculine/male self-perceptions caused by dysphoria or internalized transphobia. The contrast between these perceptions and speakers' otherwise feminine presentations may then similarly result in gay-masculine self-perceptions. These self-misgendering moments attest that the cisgender listening subject is an ideological perspective, rather than a specific cisgender individual (although these individuals often inhabit this ideological perspective). As a result, these speakers may apply the judgments of the cisgender listening subject to their own voices, seeing themselves through the marginalizing lens of hegemonic norms. In other words, that they sometimes feel upset that their voices sound gay, rather than feminine, is likely a result of the internalization of hegemonic and marginalizing gender norms.

Within the hegemonic structure of fractal-recursive interpretation of linguistic signs, indexing gay-masculinity is often indexing femininity within a context considered male, as discussed above. However, it is nonetheless the case that some linguistic variables are more salient, or well-known, as indexes of gay speech. This is the case for /s/; there is a culturally recognized metalinguistic notion of the so-called 'gay lisp' (Calder 2020b), which considers gay men's use of fronter /s/ pronunciations to sound like a 'lisp'. It is possible that when Mantis and Jellyfish hear their voices as gay-masculine, /s/ is of primary concern (consciously or unconsciously) due to its salient status as the 'most iconic' (Calder 2020b) index of gay-masculine sexuality. It is possible, then, that Mantis and Jellyfish may be more aware (consciously or unconsciously) than Alpaca and Squirrel that /s/ can index gay masculine speech (at the $n+1$st indexical order), because they may have applied this indexicality to themselves in moments of dysphoric self-misrecognition. As a result of the close connection – or perhaps underlying equivalence – of /s/ fronting's feminine and gay-masculine meanings, these speakers may also come to hold greater awareness of the nth indexical order, at which fronted /s/ indexes femininity. Thus, these speakers, for whom /s/ fronting is more salient (in its gay-masculine meaning), are seen to take up /s/ fronting (in its feminine meaning) to defend themselves against the cisgender listening subject. In other words, being concerned with sounding gay raises speakers' (conscious or unconscious) awareness of the importance of /s/ to gender norms in the hegemonic indexical field, and this awareness leads them to front /s/ to index femininity around cisgender listeners.

I consider this interpretation to provide the most unified explanation my data can provide for why Mantis and Jellyfish front /s/ with the Cis CP while Squirrel

and Alpaca do not. In reconsidering my initial prediction with regards to concern with gay-masculine misrecognition, it is clear that avoiding fronter /s/ pronunciations is unproductive for speakers. If speakers assume that their /s/ pronunciations will be interpreted through a male perceptual frame (within the fractal-recursive binary opposition at the $n+1$st indexical order), then the only indexical options with /s/ are straight man and gay man, both of which are undesirable perceptions. Instead, the best option available to speakers who are aware (consciously or unconsciously) of the structure of hegemonic indexical meanings is to front /s/, in the (conscious or unconscious) hopes that their /s/ pronunciations will be interpreted through a female perceptual frame (within the fractal-recursive binary opposition at the nth indexical order) and contribute to overall feminine perceptions.

Because this interpretation is primarily concerned with the distinction between shifting to fronter /s/ pronunciations with the Cis CP (moving towards hegemonic femininity) and not doing so, it is unable to differentiate between Alpaca, who shifted to backer /s/ pronunciations with the Cis CP (away from hegemonic femininity), and Squirrel, who did not shift. With the data I have, I am not able to provide a confident explanation for this difference. However, I suspect that the confounding factors of Alpaca's pre-existing familiarity with the Trans CP and uncertainty as to the trans status of the Cis CP likely play a role. With respect to Squirrel, in addition to not having a personal history of self-perceived gay-masculine speech, her lack of shifting may also be explained in part by her transition stage. Even if she is aware (consciously or unconsciously) that her /s/ pronunciations may effect how her gender is perceived by a cisgender listener, it is possible she is only in the habit of pronouncing /s/ in a one way, under the assumption that trans speakers may practice and learn a wider range of /s/ pronunciations throughout their transitions. Zimman (2021b) demonstrates that a multi-dimensional model of gender is needed to explain the /s/ pronunciation patterns of his gender-diverse set of speakers, including gender assigned at birth, current gender identity, and gender role – the gender the speaker currently performs on a daily basis (which for Squirrel is 'cisgender straight man', until she comes out). My data agrees with Zimman's model here, affirming the importance of considering gender role.

An additional interpretive note regards the factor of race in my data. Because the hegemonic indexical field for /s/ is based upon specifically white gender norms, it is possible that Mantis' shift also worked to protect her from the judgments of the white listening subject. A number of studies of race and style-shifting have found that Black speakers shift towards hegemonically white normative speech with white listeners (eg. Rickford and Mcnair-Knox 1994). It is possible that Mantis, as an Asian-American, engages in a similar practice here. This may also explain why Mantis' shift was the most drastic. For her, the contrast between the Trans CP and the Cis CP may be multiplied by the the racial contrast between the CPs; the Trans CP is Chinese, and the Cis CP is white. Therefore, with the Cis CP, Mantis may have been reacting to the possible presence of both the cisgender and the white listening subjects, thus doubly impelled to shift towards hegemonic norms. This could be seen as an example of the multiplicative oppressions experienced by multiply-marginalized speakers, an example of sociolinguistic intersectional-

ity (miles-hercules 2022). In my literature review, I was unfortunately unable to find research which discusses Asian-Americans' /s/ pronunciations; research in this area could provide further insights here.

Above, I consider the primary factor of personal history of gay-masculine self-perception, as well as secondary factors of transition stage/gender role and race, to explain whether speakers shift. Under this interpretation, my data provides evidence for the relevance of the theory of cisgender white listening subject. It highlights that indexical meanings are interlinked, and that speakers make use of complex indexical understandings to protect themselves from the judgments of this listening subject. In addition to its relevance to transfem thinking about the voice and being heard (Heffernan 2023) and its inhabitation by real cisgender listeners (Jessee and Calder 2025), it is clear that the influence of the cisgender white listening subject does, for Mantis and Jellyfish, manifest in /s/ shifting. This study thus supports the usefulness of listening subjects, indexical inventories, and indexical orders as a theorization of the mediation between perception of speech and production of speech, especially for trans speakers.

4.3 Further interpretations: Relationship with Hegemonic Norms and the Ability to Shift

In my initial model, in addition to the issue of gay-male misrecognition, which has been reinterpreted above, I made predictions regarding speakers' relationship with hegemonic femininity, as well as concern with cisgender perceptions. As discussed in the results section, all four speakers expressed these feelings in their own ways. They all noted how they might change how they speak around cisgender people, and mentioned ways in which cisgender perceptions pressure them to conform or make them feel dysphoric. All four speakers expressed nuanced relationships with femininity and described how the process of transition has brought them to their own form of femininity, which may not adhere to hegemonic norms. Because these factors were primarily similar across all of my speakers, I do not find them to determine whether speakers shift /s/. This emphasizes the powerful position of the cisgender listening subject; under my interpretation, speakers prioritize protecting themselves against cisgender perceptions, and shift /s/ if they have sufficient (conscious or unconscious) awareness of the importance of /s/ within hegemonic gendered speech norms.

Although it is possible that Squirrel and Alpaca's lack of shifts could be read through the lens of resistance, where the risk of negative cisgender perceptions is accepted in exchange for the freedom of casting off hegemonic norms, I am hesitant to make this conclusion. As mentioned above, all speakers express discontent with hegemonic norms and cisgender perceptions; I have no reason to believe that some speakers resist hegemonic femininity more than others, and it is not my place as a researcher to make these judgments. Additionally, I do not wish to imply that Mantis and Jellyfish's shifts could not be considered resistance; they resist the

cisgender listening subject by taking up /s/ as a hegemonic indexical resource, refusing to allow themselves to be misrecognized. Furthermore, it is difficult to determine whether speakers are, on the whole, moving their speech towards or away from hegemonic norms, as I have only investigated /s/, one small part of these norms. Future research which considers /s/ as one element of a constellation of indexically-relevant sites of speech variation, including, for example, overall pitch (f0), intonation, voice quality, or discourse markers, could provide greater clarity here. For the moment, I am unable to conclude whether my speakers should be said to be resisting, and hesitate to make the judgments which may be required to make this sort of determination.

Nonetheless, I encourage future research to investigate this topic with nuance, as I do believe trans and otherwise marginalized speakers implement social-indexical resistance. Barrett and Hall (2024) mention the ways in which marginalized groups can take advantage of misrecognition as semiotic resistance, or can recontextualize hegemonic indexicals into subverting contexts (termed indexical disjuncture). Sociolinguistic research which investigates more closely the connection between speakers' expressed views of how they resist normativity and their language use could further illuminate this balance between self-protection and defiance of norms, both of which can be seen as resistance in their own ways.

An additional factor which may be relevant to my data concerns intentionality and accessibility of shifting. In my interviews, it was clear that speakers did not feel as though the gendered sound of their voice was fully under their own control, and that gaining this control may require effort and practice. As mentioned above, Jellyfish characterizes 'talking like a gay guy' as a type of 'default'ing. She uses the metaphor of rolling dice to describe her relationship with the various voices she feels she has access to,

> "It's kind of rolling the dice what voice comes out on a certain day, and sometimes the faggcent comes back"

and considers 'talking like a boy' to be her neutral state:

> "at a certain point, I like stop performing and sort of default back to like my neutral state which is still just like talking like a boy, which is fine, like I don't really feel much pressure to change that."

In these ways, Jellyfish is not in complete control of her voice. On the other hand, she also expresses that she is proud of her ability to style-shift, and that this grants her freedom:

> "I'm pretty proud of my voice, and like, um, being able to more or less manipulate it how I want, um, and, I feel like it gives you a lot of freedom in like how you move through the world if you can like talk in different ways to different people."

The overall picture that is drawn here is that Jellyfish considers style-shifting, and controlling her voice in general, to be a learned skill, rather than an inherent

ability, and attributes times when her voice is not the way she would like it to be to the lack of complete control.

My other speakers view their relationships with their voices similarly. Squirrel mentions that her voice is "not the way [she] would want it to be if [she] could like, pick it" when describing her voice dysphoria. Alpaca mentions that she learned "the exercise to raise your larynx ... and then [she] found [her]self sort of subconsciously doing it" over time; here, the ability to style-shift easily takes time to develop, whether through conscious effort (in Alpaca's case), or unconscious changes in habit. For Mantis, the contrast between her discussions of struggles with voice dysphoria and success in passing with her voice similarly draw a picture of slow changes in linguistic habits, rather than immediate control over linguistic variation.

I find this tendency notable because it is somewhat out of alignment with the general assumption of modern frameworks of social-indexical linguistics, that speakers' behavior can, and should, always be interpreted as intentional and aligned with their indexical goals. In the interpretation of my results put forth above, I consider the insights my speakers provide about their partial control over their voices helpful. From my data, It seems that all of my speakers intend to defend themselves against the judgment of the cisgender listening subject. However, in order to do so with /s/, it seems that the indexicalities of this sound need to carry a certain level of salience (as a result of concerns with sounding gay-masculine), and that a certain length of time since coming out as trans may be necessary (as shown by Squirrel's case). I therefore encourage future sociolinguistics research to take up the issue of intentionality more carefully, as I attempt to do here. If a first attempt to explain speakers' behavior as aligned with their identity and goals does not provide sufficient answers, it is possible that forcing this assumption may be problematic.

Conclusion

In conducting a comparison of transfeminine speakers' /s/ pronunciations with a cis listener and a trans listener, this thesis sheds light on the ways that marginalized individuals navigate norms which exclude and restrict them. The field of language and gender is well-suited to this investigation, as it provides in-depth understandings of linguistic gender norms through close analysis of perception and production of social meanings through language use. However, research on trans speakers has primarily taken two approaches to hegemonic gender norms; it has either compared trans speakers against cis speakers and characterized their differences from these norms, or analyzed the unique patterns of language use constructed within trans-safe communities. By investigating directly the effect that the presence of cisgender listeners has on transfeminine speech, I fill this gap, answering two theoretical questions present in the existing research on marginalized speakers.

First, in response to a growing body of research on privileged listening subjects – ideological positions which apply hegemonic, marginalizing understandings in the interpretation of sociolinguistic signs – I respond to the question of how speakers react to the presence of a listener who may inhabit this sort of privileged listening subject position in actual linguistic practice. Second, I take up the question of how speakers model their listeners' indexical inventories, aiming to understand the mediation between linguistic perception and linguistic production. I find that two speakers move towards fronter /s/ pronunciations with the Cis CP and two do not, and interpret the implications of my findings for these questions.

With respect to the first question, although I am unable to prove with certainty that my speakers' shifts towards fronter /s/ with the Cis CP are not simply a matter of accommodation, I find that considering their shifts as reactions to a perceived presence of the cisgender listening subject produces a stronger interpretation. In combination with my finding that my speakers construct the cisgender listening subject within their metalinguistic commentaries (similarly to Heffernan 2023), I argue that transfems' concept of the cisgender listening subject influences their actual linguistic practice. I consider the speakers who shift towards fronter /s/ pronunciations to be protecting themselves from the possibility of misgendering judgments by moving into greater alignment with /s/ pronunciations hegemonically considered feminine.

In considering why only two of my four speakers shift this way, although I am similarly unable to explain this difference with certainty, I notice that the speakers

who fronted /s/ with the Cis CP discuss a history of gay-masculine self-perceptions of the voice. The indexical link present in the hegemonic indexical field from fronted /s/ pronunciations to gay men's speech is highly salient (Calder 2020b). I find it likely that /s/ fronting, in both its gay-masculine and its feminine meanings, is more salient to speakers concerned with sounding gay-masculine. In this case, they take up /s/ fronting to defend themselves from the perceptions of the cisgender listener in the hopes that it will not be misrecognized as a gay-masculine index.

With respect to the second question posed above, I conclude that marginalized speakers model the indexical fields of privileged listeners through complex and complete models of the hegemonic indexical field. In my interpretation, speakers' model the hegemonic indexical field of /s/ in line with the fractal-recursive indexical order model put forth by Calder (2019a); /s/ fronting enters my speakers' (conscious or unconscious) awareness through its salient meaning of gay-masculinity at the $n+1$st indexical order, but this awareness also includes the binary-gendered indexical link to femininity on the nth indexical order. As an example of the close interdependence of indexical orders (see Silverstein 2003), it can be seen that marginalized speakers come to a complete (conscious or unconscious) understanding of the hegemonic indexical field, and respond in consideration of this full structure.

In light of the theory of indexical misrecognition put forth by Barrett and Hall (2024), I see that marginalized speakers can employ generalized understandings of hegemonic meanings – their understandings of the ideological positions of cisgender and otherwise privileged listening subjects – to model the indexical inventories of specific privileged listeners. Here, speakers act under the assumption that these listeners will take up a privileged listening subject position and apply the hegemonic field to all speakers – an act of indexical presumption – and speakers protect themselves from indexical misrecognition accordingly.

Taken as a whole, my study hopes to simultaneously highlight the difficulty of operating within a system of exclusive and marginalizing norms, and the freedom speakers can build to navigate those norms as they choose. As my participant Jellyfish says,

> "Fundamentally, queerness should change how you move through the world. I think that's a good thing, I think like having control of how you're perceived and how you interact with the world and thinking about that more is something that everyone should do, and my transness and my lesbianism are like vectors to which I do that."

5.4 Limitations and Directions for Future Research

A primary limitation of this study relates to its lack of raciolinguistic focus. As a white researcher working with mostly white participants and drawing on a mostly white-centric research literature, I acknowledge the shortcoming that I perpetuate: the whiteness of trans studies, queer studies, and sociolinguistics. I have

attempted to point out the racial dynamics in my data as I can see them, and to contextualize the whiteness of the norms I investigate, to mitigate this issue and situate my research within its specific racial context. Given the whiteness of the hegemonic gender norms around /s/ (A. J. Steele 2022; J. Calder and King 2022), it would be particularly illuminating to investigate BIPoC transfeminine speakers' /s/ pronunciations in more depth. A study with a similar methodology as mine but which specifically focuses on this population may find that race of the perceiver is equally relevant as cisgender status, and that the influence of the white cisgender listening subject would exert multiplicative oppressive effects, an example of sociolinguistic intersectionality (see miles-hercules 2022), as hinted at by the behavior of my Asian-American speaker, Mantis. I encourage future research to work outside white speaker and perceiver populations, and hope to be able to work with queer and trans BIPoC sociolinguists and speakers to look more closely at the factor of race in the future.

The other primary limitations of this study pertain to scale and scope. As mentioned in my discussion chapter, this study could greatly benefit from more angles of analysis in several ways. To name two, expanding to include multiple linguistic indexes of gender alongside /s/ would provide much-needed context about how speakers engage with hegemonic norms more broadly, and zooming in to the discourse level to inspect how /s/ was shifted within each individual conversation based on topic, stance, and mood would allow a more nuanced understanding of the moment-to-moment meaning-in-context of /s/ as it develops and shifts, which is lost in the aggregated understanding I present. Further research would also benefit significantly from a larger number of participants, at which a greater diversity of participants and conversation partners could reveal broader patterns and reduce the possible confounds introduced by any idiosyncrasies of the individual characteristics of the two conversation partners I have. This would also allow for a quantitative analysis of predictive factors of /s/ shifting, operationalizing more precisely the attitudes, concerns, and identities I consider qualitatively here.

Appendix A

Survey

The survey given to participants is reproduced below:

Survey

Transfeminine Experiences and Interactional Styles

This survey will ask you questions about your identity, experiences, and background. Information you provide here will help me to consider a broader picture of you as I analyze your linguistic behavior. If you have any questions for me, feel free to ask at any point. All questions are optional, and you are encouraged to skip any questions which may make you feel uncomfortable, and to answer in a level of detail most comfortable for you. Additionally, keep in mind the risk of identification associated with sharing recognizable information, especially that which may be unique to you.

One- or two-word answers are sufficient for these questions.

For all questions, if none of the options provided fit you, you are encouraged to write something else, choose not to answer, or indicate that you would like to elaborate in our recorded conversation by writing "elaborate".

Pseudonym:
Pronouns:

Gender Identity
What (if any) gender identity label(s) do you identify with? (e.g. Woman, Girl, Nonbinary, Two-Spirit, Genderfluid, Agender, Demigender, etc.)

For the following two questions, feel free to answer "fluid" or "elaborate" if a numerical rating is not fitting for you.
On a scale of 0 (not feminine) to 100 (very feminine), how much do you generally identify with femininity?

On a scale of 0 (not masculine) to 100 (very masculine), how much do you generally identify with masculinity?

Sexuality
What (if any) sexuality label(s) do you identify with? (e.g. Lesbian, Bisexual, Pan-

sexual, Queer, Straight, Gay, Asexual, Aromantic, or Ace Spectrum, etc.)

Other Aspects of Identity
What (if any) racial/ethnic identity labels do you identify with? (e.g. Native American, Black, East Asian, South Asian, Latina/e/o, North African/Middle Eastern, White, etc.)

Do you have, or have you had, any disabilities which may relate to speech? (y/n; if yes, briefly describe)

Linguistic Background
Do you speak any languages other than English? (y/n; if yes, list)

Have there been periods of more than a year where you primarily spoke a language other than English? (y/n; if yes, briefly explain)

Where did you grow up?

(a) Outside the US

(b) Multiple Locations

(c) West US

(d) Midwest US

(e) Northeast US

(f) South US

The Body [this section only shown to trans participants]
Are you currently on hormone replacement therapy (HRT)? (y/n)

If so, approximately how long have you been on HRT? (number of years)

Have you received voice feminization surgery? (y/n)

Experience [this section only shown to trans participants]
Have you received voice feminization therapy? (y/n)

Have you received other speech therapy? (y/n)

Appendix B

Interview Questions

In order to facilitate free-flowing conversation in my interviews with my participants, I asked different questions depending on the topics that were brought up and issues which I felt may have been relevant to each speaker based on their survey responses or comments made to me during the recruitment process. This appendix is therefore separated into sections for each speaker, each of which containing a list of all of the questions I asked them in our interview.

B.1 Speaker 1: Mantis

- "I wanna ask a little bit about like your understanding of like, or your conceptualization of your gender? Um, and maybe also sort of the the question of how that's developed over time and like how you came to sort of identify the way you do now and what that sort of process has been like."
- "Is it sort of like an 'I'm faking if I don't ...?"
- "How do you think that [your sexuality] sort of interacted with the question of whether you were trans?"
- "And had you been attracted to men in the past?"
- "Like he didn't want you to be a woman?"
- "Do you think if you were cis you would feel able to like, and want to inhabit like a butch, sort of role in a lesbian, women loving women relationship? Does it feel like you're not really able to?"
- "What has been your experience with your own voice throughout transitioning and what's it like now?"
- "When you did sort of worry about [your voice], what was the worry like? Was it just sort of, not feminine enough? Or...?"

- "Do you think that as you sort of became less worried about like, being mistaken for a gay man through your voice do you think that changed the way that you tried to speak and present yourself through your voice? Or do you think it was more that you like felt less concerned with how it already was?"
- "Do you feel like you notice yourself changing the way you speak depending on who's around and stuff very much?"
- "Do you feel like you have to speak more femininely around cis people in general?"

B.2 Speaker 2: Alpaca

- "Wanted to just sort of ask a little bit about the sort of trajectory of your transition through your life, and like how that went and like, so like when you first started to realize, and then like once you started making changes, and what kind of changes you've made?"
- "Okay so how much later was that [moment of realization that she is trans], was that like, after you graduated, or...?"
- "Does it feel to you like [using it/its pronouns] is sort of like also a feminine thing, or does that sort of stand outside of femininity for you?"
- "Did you do any voice training?"
- "Do you feel like you notice yourself speaking differently around different people?"
- "You said that 'lesbian' was the sexuality label, do you feel like you ever could imagine presenting your voice as sounding lesbian in any sort of way? Do you think that even is something that seems possible?"
- "Do you ever want to sound specifically like a trans woman?"
- "If you were to guess, do you think that like, at this point in time, when you're just like meeting somebody new, they're likely to assume that you're a cis woman? And does that make it that coming out to that person would be more of an 'I'm trans' than 'I'm a woman'?"
- "If you're willing to talk about it, do you feel like you have much voice dysphoria?"
- "If you think of aspects of the voice, what's the number one thing you think of as far as gender? Do you know of any others, are there any others that come to mind?"
- "You said your speech therapy was about the 'th' sound?"

B.3 Speaker 3: Squirrel

- "You said to me that you're fairly early transition. So, what's the timeline for that? What brought you to first realize?"
- "So who all knows [that you're trans]?"
- "Oh yeah you were talking a little bit about a situation with your parents with [the Trans CP] I think, what was that?"
- "I did want to talk a little bit about what brought you to sort of realize that you were trans, and when that happened"
- "The sort of moment where you were like, okay I actually am, when was that?"
- "It seems like you have kind of a bit of a plan as far as like starting to, would you say it's like starting to present fem?"
- "Do you have like a sort of sexuality label that you use? How do you feel like that sort of ties into your gender?"
- "With this, sort of, vague plan that you have, how do you think the voice, how you speak, is going to factor into that?"
- "So once you sort of begin to present fem and come out to people at reed, do you think part of that process will involve any sort of like attempts to change your voice? Are you gonna like do some sort of like voice training?"
- "Do you feel like your voice is feminine at all now?"
- "[you talk differently around] different people, you said. How would you categorize that or like what kind of people. Is it personal, or is it more sort of general?"
- "Do you think that you speak differently with other trans people, other queer people, versus cis people?"
- "Is there any sort of, um, fear or uncertainty about using certain ways of speaking around certain people at all for you? Or just like discomfort maybe?"
- "Do you feel like you have a voice that you use with strangers that's sort of like different? You've probably heard the term like customer service voice? Do you feel like you have something like that?"
- "Like if you're ordering at a restaurant, maybe. Do you think you'd be speaking differently in that sort of scenario?"

- "We've been talking about sort of like, sounding fem. Do you feel like there is at all, do you think that you might be able to or might aim to speak in a way that sort of sounds in a way that might be aligned with your sexuality as well?"
- "If you were a cisgender woman, is there like a way that you would imagine that you'd be? I guess what I mean by that is like, do you think, cause like I think maybe some trans people probably feel as though like, um, if they had more sort of latitude they might be able to, say, be butch or something like that. Where there might not be as much access to that. Do you feel that kind of thing?"

B.4 Speaker 4: Jellyfish

- "So I think that the first thing I wanna ask is maybe just for you to like tell me a little bit about your sort of understanding of your gender and your sort of relationship with gender as a thing?"
- "How you came to realizing that [you were trans] and sort of what the journey has been?"
- "As I've been talking to various people, there's like a bit of a maybe like a flip, where like sort of like being out goes from 'you know that I'm a woman' to 'you know that I'm trans' so it seems like you sort of had that flip happen?"
- "So that sort of like, being perceived mostly or assumed to be a cis woman, do you think that was like more or less what you wanted?"
- "Trying to not be like other trans girls or whatever?"
- "You said [you identify with the term] 'dyke', do you think that you sort of project a lesbian identity, do you think that people generally know?"
- "First of all, like what's your sort of relationship with your voice now? And then like, how has that evolved over time?"
- "So for you maybe was it sort of like there was a period of time where you were sort of like actively paying a lot of attention to what you were doing with your voice and like trying to make changes?"
- "The thing you mentioned about like the idea of being concerned about sounding like a gay guy is one of the sort of questions that I'm interested in... From the perspective of the voice, do you feel like that comes up as a concern, and do you know if there's any things that you maybe do to like try to address that?"
- "If you use like sort of like, ... a voice that sounds like a boy do you feel like that could read as sort of butch or whatever and like that's fine?"

- “Do you think that the sort of like with close friends circle, to what extent does that extend to like just like with other trans people in general?”
- “When you think about the gender of a voice, what other things [than voice pitch] maybe come to mind?”
- “Have you like researched [gendered aspects of the voices]?”

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